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# A generic binding pocket for small molecule $I_{Ks}$ activators at the extracellular inter-subunit interface of KCNQ1 and KCNE1 channel complexes

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## Abstract

The cardiac $I_{Ks}$ ion channel comprises KCNQ1, calmodulin, and KCNE1 in a dodecameric complex which provides a repolarizing current reserve at higher heart rates and protects from arrhythmia syndromes that cause fainting and sudden death. Pharmacological activators of $I_{Ks}$ are therefore of interest both scientifically and therapeutically for treatment of $I_{Ks}$ loss-of-function disorders. One group of chemical activators are only active in the presence of the accessory KCNE1 subunit and here we investigate this phenomenon using molecular modeling techniques and mutagenesis scanning in mammalian cells. A generalized activator binding pocket is formed extracellularly by KCNE1, the domain-swapped S1 helices of one KCNQ1 subunit and the pore/turret region made up of two other KCNQ1 subunits. A few residues, including K41, A44 and Y46 in KCNE1, W323 in the KCNQ1 pore, and Y148 in the KCNQ1 S1 domain, appear critical for the binding of structurally diverse molecules, but in addition, molecular modeling studies suggest that induced fit by structurally different molecules underlies the generalized nature of the binding pocket. Activation of $I_{Ks}$ is enhanced by stabilization of the KCNQ1-S1/KCNE1/pore complex, which ultimately slows deactivation of the current, and promotes outward current summation at higher pulse rates. Our results provide a mechanistic explanation of enhanced $I_{Ks}$ currents by these activator compounds and provide a map for future design of more potent therapeutically useful molecules.

## Significance Statement

Combined, KCNQ1 and KCNE1 subunits generate the $I_{Ks}$ current. Activating $I_{Ks}$ has been identified as a promising therapeutic strategy to treat arrhythmogenesis resulting from delayed repolarization. In this study, we uncovered a common drug-induced binding site accessed by two structurally diverse $I_{Ks}$ activators, mefenamic acid and DIDS. Located in the extracellular interface where KCNE1 interacts with KCNQ1, we propose that binding of drugs to this location impairs channel closing and leads to enhanced current activation. This is shown to be particularly beneficial at higher pulse rates and explains how such drugs may make an important contribution to the electrical repolarization reserve in the heart.

### eLife assessment

By combining electrophysiological analysis of mutant channels and molecular dynamics simulations, this **important** study identifies a common binding site for two structurally distinct activators of KCNQ1-KCNE1 channels. The findings represent an **important** advance for the field, with **convincing** functional data to support it. The simulations are still partially **incomplete** and would benefit from additional data and discussion. The work will be of interest to those studying the binding of small molecule drugs to membrane protein complexes.

# Introduction

Potassium ion ($K^+$) channel activators are important compounds in human health, as partial or complete loss of function of many $K^+$ channels may lead to inherited or acquired diseases that have significant morbidity and mortality. The delayed cardiac rectifier potassium current, $I_{Ks}$, plays an important role, especially at high heart rates, in the physiological shortening of the cardiac action potential (Sanguinetti et al., 1996). Unsurprisingly, mutations in both KCNQ1 (α-subunit) and KCNE1 (β-subunit), which when paired together give rise to the $I_{Ks}$ current (Barhanin et al., 1996; Sanguinetti et al., 1996; Bendahhou et al., 2005), have been implicated in cardiac arrhythmia syndromes such as long QT syndrome (LQTS) and atrial fibrillation (Jervell and Lange-Nielsen, 1957; Wang et al., 1996; Chen et al., 2003; Eldstrom and Fedida, 2011; Olesen et al., 2014). With nearly all mutations seen in LQTS patients identified as loss-of-function and 50% of those loss-of-function mutations identified in the KCNQ1 subunit (Hedley et al., 2009; Ackerman et al., 2011), enhancing and activating $I_{Ks}$ currents has long been suggested as a promising therapeutic approach for treating LQTS. It is a curiosity of $I_{Ks}$ channels that the known activators fall into two general groupings: those that work best on the α-subunit, KCNQ1, alone, including zinc pyrithione, L-364,373, and one of the most studied activators, ML277 (Mattmann et al., 2012; Yu et al., 2013; Xu et al., 2015; Eldstrom et al., 2021); and those that work best in the presence of the auxiliary β-subunit, KCNE1, compounds like mefenamic acid and DIDS (4,4'-diisothiocyano-2,2'-stilbene disulfonic acid) (Abitbol et al., 1999; Wang et al., 2020).

Using cryo-EM, we recently visualized the binding of ML277 deep in the central core of KCNQ1 channels in a pocket lined inferiorly by the S4-S5 linker, laterally by the S5 and S6 helices of two separate subunits, and above by pore domain residues (Willegems et al., 2022). The location of the binding pocket and its structural inter-relationships help to explain the underlying mechanism of action of ML277, its specificity for KCNQ1, and the lack of efficacy due to steric hindrance in the presence of a β-subunit. However, it is known neither where activators of $I_{Ks}$ bind that *require* the presence of the β-subunit, such as phenylboronic acid (Mruk and Kobertz, 2009), hexachlorophene, stilbenes such as DIDS and SITS (4-acetamido-4'-isothiocyanatostilbene-2,2'-disulfonic acid), and diclofenac acid derivatives such as mefenamic acid (Abitbol et al., 1999; Zheng et al., 2012; Wang et al., 2020), nor how they mediate their activator action. We have some clues for DIDS and mefenamic acid that their binding sites are not in the central channel core, as is the case for ML277, but depend on KCNE1 β-subunit residues at the extracellular surface of the channel, residues 39-43 in

KCNE1 for DIDS (Abitbol et al., 1999) and K41 for mefenamic acid (Wang et al., 2020). Our recent cysteine scanning data revealed that although other extracellular KCNE1 residues in the same region to varying degrees impacted the effect of mefenamic acid, only the K41C mutation completely abolished mefenamic acid effect up to a concentration of 1 mM (Wang et al., 2020). Previous cross-linking studies have identified key interactions between this extracellular region of KCNE1 and the S1 and S6 transmembrane segments of KCNQ1 (Xu et al., 2008; Chung et al., 2009), suggesting that residues in either of these regions could also provide important clues to explain mefenamic acid's mechanism of action.

To further explore the dependence of residues in KCNE1 and those in adjacent KCNQ1 sites on binding of mefenamic acid to $I_{Ks}$, we first examined the role of K41C in preventing the drug effect. Docking in combination with molecular dynamics (MD) simulations of mefenamic acid binding to $I_{Ks}$ followed by mutagenesis were used to map out critical KCNE1 and KCNQ1 residues. Further, we expanded on the idea that the stilbene, DIDS (Abitbol et al., 1999), which is structurally quite different from mefenamic acid, shares a common binding site. Our results showed that both compounds bind in the same general region formed by elements of the pore and S6 domains of KCNQ1 and the near extracellular region of KCNE1, but depend on different critical residues for their binding stability. Exposure of the channel complex to either compound induces subtle structural changes that subsequently stabilize the conformation of the S1/outer pore/S6 of KCNQ1 and slow the $I_{Ks}$ deactivation gating kinetics. The results suggest the existence of a common drug-induced binding site and a mechanism of action for small molecule $I_{Ks}$ activators which is distinct from that of specific compounds that activate KCNQ1 alone.

# Results

## The mefenamic acid binding site on the KCNQ1/KCNE1 complex

Exposure of wild type $I_{Ks}$ complexes (4:4 ratio; WT EQ) to 100 µM mefenamic acid transforms the slowly activating $I_{Ks}$ current into one with an almost linear waveform and completely inhibits tail current decay at -40 mV (Figure 1A) (Abitbol et al., 1999; Unsöld et al., 2000; Wang et al., 2020). G-V relations obtained from peak initial tail currents show that 100 µM mefenamic acid hyperpolarizes the G-V ($\Delta V_{1/2}$ = -105.7 mV, Figure 1B, 1C) and decreases the slope (control *k* = 19.4 mV, Mef *k* = 41.3 mV, Table 1). We previously showed that introduction of a cysteine mutation at residue K41 in all four KCNE1 subunits (4:4 ratio of mutant K41C-KCNE1 to KCNQ1; K41C-EQ) itself had only a minor effect on the G-V, but prevented changes to currents and the G-V relationship on exposure to 100 µM or 1 mM mefenamic acid (Figure 1B, 1C and Table 1) (Wang et al., 2020). The data suggest that residue K41 in KCNE1 is critical to the action of mefenamic acid, but give no information why K41 is so important, and prompted us to question how this residue and other adjacent residues in KCNE1, and those nearby on KCNQ1 acted together to form a binding pocket for mefenamic acid.

### Table 1.

Mean $V_{1/2}$ of activation (mV) and slope values (*k*-factor, mV) in the absence and presence of mefenamic acid for fully saturated $I_{Ks}$ channel complexes. A statistical difference in $V_{1/2}$ compared to control is shown as p-value, determined using an unpaired t-test. NS denotes not significant. SEM is denoted by ±.

| | **Control** | | | **100 µM or 1 mM Mefenamic Acid *** | | | **p-value** |
|---|---|---|---|---|---|---|---|
| | $V_{1/2}$ | *k-factor* | n | $V_{1/2}$ | *k-factor* | n | |
| **Wildtype EQ** | 25.4 ± 2.4 | 19.4 ± 1.2 | 6 | -80.3 ± 4.1 | 41.3 ± 8.4 | 3 | <0.0001 |
| **K41C-EQ** | 15.2 ± 1.1 | 18.4 ± 1.7 | 4 | 11.4 ± 1.0 | 19.4 ± 0.8 | 4 | <0.05 |
| | | | | 16.7 ± 2.0 | 19.8 ± 1.4 | 3 | NS |
| **L42C-EQ** | 68.9 ± 1.5 | 21.5 ± 3.7 | 3 | 31.8 ± 0.4 | 14.8 ± 4.27 | 3 | <0.01 |
| **E43C-EQ** | 70.8 ± 1.8 | 27.0 ± 2.8 | 4 | 46.4±2.1 | 29.1 ± 1.7 | 4 | <0.01 |
| **A44C-EQ** | 4.1 ± 1.8 | 17.6 ±1.4 | 4 | -5.6 ± 2.7 | 18.3 ± 2.0 | 4 | <0.05 |
| **Y46A-EQ**** | 76.4 ± 1.5 | 57.3 ± 1.6 | 3 | -29.8 ± 4.1 | 18.9 ± 3.1 | 3 | <0.05 |
| **EQ-W323A** | 47.8 ± 2.7 | 23.7 ± 2.2 | 4 | 33.7 ± 1.2 | 28.5 ± 3.1 | 3 | <0.05 |
| **EQ-W323C** | 54.0 ± 2.1 | 22.2 ± 2.1 | 4 | 27.5 ± 3.3 | 25.6 ± 1.8 | 4 | <0.05 |
| **EQ-V324A** | 34.4 ± 2.3 | 20.7 ± 1.1 | 6 | 15.5 ± 2.3 | 27.9 ± 1.6 | 5 | <0.05 |
| **EQ-V324W** | 41.0 ± 2.4 | 18.4 ± 0.3 | 4 | 27.3 ±6.0 | 25.5 ± 1.3 | 4 | NS |
| **EQ-Q147C** | 63.9 ± 5.8 | 25.4 ± 2.0 | 4 | 26.3 ± 5.7 | 32.7 ± 1.7 | 4 | <0.05 |
| **EQ-Y148C** | 36.8 ± 0.3 | 20.3 ± 0.4 | 4 | 17.5 ± 4.0 | 36.7 ± 3.7 | 4 | <0.05 |

* For K41C-EQ, the concentration of mefenamic acid used was either 100 µM (upper row values) or 1 mM (lower row values). For all other constructs, 100 µM mefenamic acid was used.
** An estimation of the activation $V_{1/2}$ was calculated from a right shifted, non-saturating GV curve.

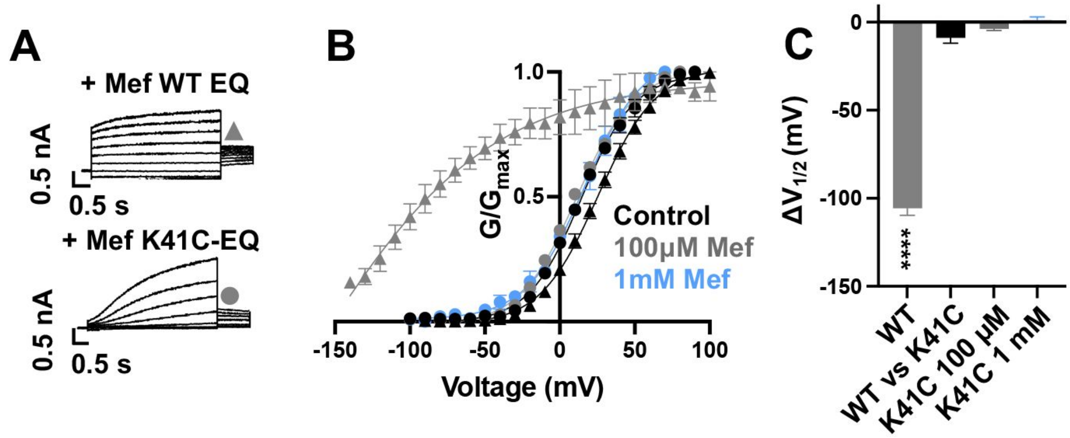


**Figure 1.**

**K41C-KCNE1 mutants prevent the agonist effect of mefenamic acid.**

(A) Current traces of WT EQ (top) and K41C-EQ (bottom) in the presence of 100 μM mefenamic acid (Mef). A 4 s protocol was used with pulses from -150 mV or higher to +100 mV, in 10 mV steps, followed by a repolarization step to -40 mV for 1 s. Holding potential and interpulse interval were -80 mV and 15 s, respectively.

(B) G-V plots obtained from WT EQ tail currents (triangles) and K41C-EQ (circles) in the absence (control: black) and presence of Mef (100 μM: grey; 1 mM: light blue). Boltzmann fits were: WT EQ control (n = 6): $V_{1/2}$ = 25.4 mV, *k* = 19.4 mV; WT EQ 100 μM Mef (n = 3): $V_{1/2}$ = -80.3 mV, *k* = 41.3 mV; K41C-EQ control (n = 4): $V_{1/2}$ = 15.2 mV, *k* = 18.4 mV; K41C-EQ 100 μM Mef (n = 4): $V_{1/2}$ = 11.4 mV, *k* = 19.4 mV; and K41C-EQ 1 mM Mef (n = 3): $V_{1/2}$ = 16.7 mV, *k* = 19.8 mV.

(C) Summary plot of $V_{1/2}$ change ($\Delta V_{1/2}$) for WT EQ in the presence of 100 μM mefenamic acid, WT EQ vs K41C-EQ in control and K41C-EQ in the presence of 100 μM and 1 mM mefenamic acid. **** denotes a significant $\Delta V_{1/2}$ compared to control where $p<0.0001$.

## Mefenamic acid binding site predicted by molecular modeling

Initially, to visualise potential drug binding sites and understand how mutation of KCNE1 and KCNQ1 residues might prevent drug action, *in silico* experiments of drug docking with subsequent MD simulations were performed on a model of $I_{Ks}$ channels. The $I_{Ks}$ model was constructed based on the recent cryo-EM structure of KCNQ1-KCNE3, which is thought to represent the activated-open state of the channel complex (Sun and MacKinnon, 2020). Taking into consideration the sequence similarity of KCNE1 and KCNE3 subunits in their transmembrane segments, it has been suggested that the main interface of these subunits with KCNQ1 is preserved in this region. Our initial data indicated that the extracellular residues of KCNE1 are involved in the action of $I_{Ks}$ activators, so we constructed a model where external KCNE3 residues R53-Y58 were substituted with homologous KCNE1 residues, D39-A44 (Figure 2A). The resulting 4:4 $I_{Ks}$ channel complex was termed pseudo-KCNE1-KCNQ1, ps-$I_{Ks}$, and Figure 2B shows the essential elements of the ps-$I_{Ks}$ subunits which form extracellular interface of KCNQ1 and KCNE and served as a basis for drug docking and MD simulations. Details of the docking procedure are described in the Methods section and schematically summarized in Figure 2-figure supplement 1.

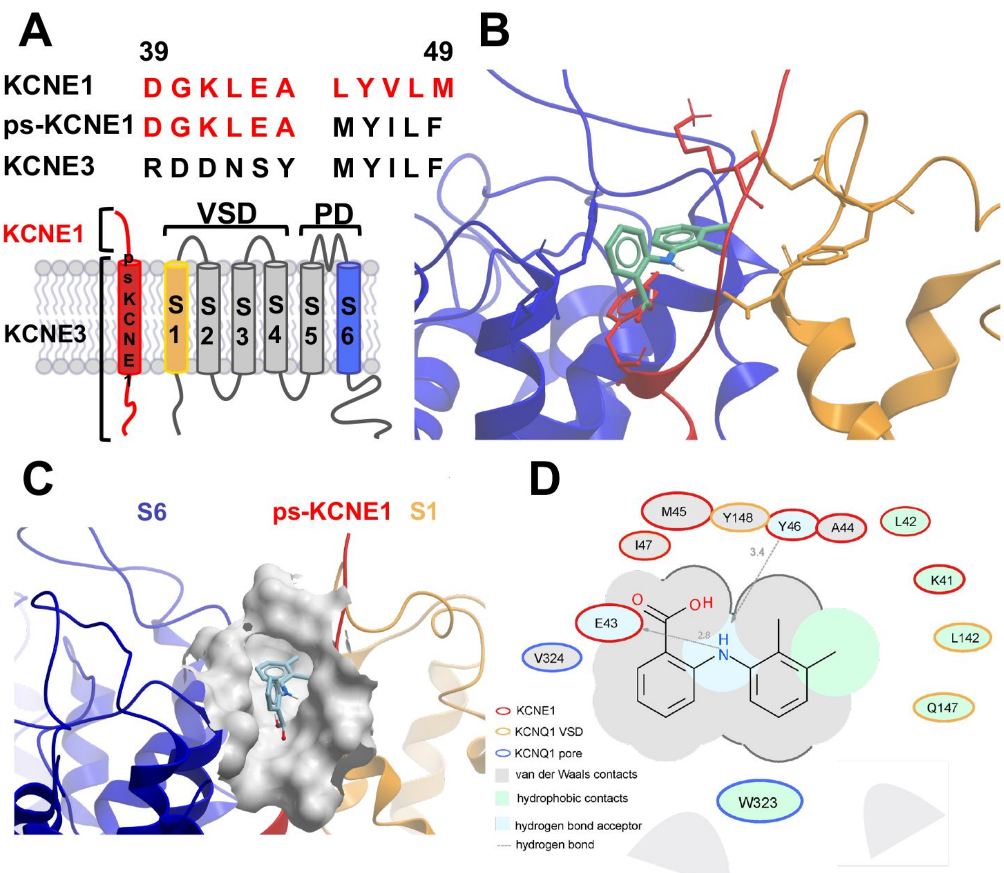


**Figure 2.**

**MD prediction of mefenamic acid binding site in the ps-$I_{Ks}$ model.**

(A) Pseudo-KCNE1 (ps-KCNE1) used to predict Mef binding site. Extracellular residues of KCNE1 (top), ps-KCNE1 (middle) and KCNE3 (bottom). Below, cartoon topology of the single transmembrane ps-KCNE1 β-subunit and the six transmembrane KCNQ1 α-subunit. S1-S4 transmembrane segments form the voltage sensor domain and S5-S6 form the pore domain.

(B) Predicted binding pose of Mef (green) in external region of the ps-$I_{Ks}$ channel complex (side view). The residues of the pore domain are coloured in blue, ps-KCNE1 subunit in red, and the voltage-sensor domain of a neighbouring subunit is presented in yellow.

(C) Mef binding pocket in space-fill formed by external S1 (yellow) and S6 (blue) transmembrane domains of KCNQ1 and extracellular region of the ps-KCNE1 subunit (red).

(D) Ligand interaction map of Mef with ps-$I_{Ks}$. Size of residue ellipse is proportional to the strength of the contact. The distance between the residue label and ligand represents proximity. Grey parabolas represent accessible surface for large areas. The 2D diagram was generated by ICM pro software with cut-off values for hydrophobic contacts of 4.5 Å and hydrogen bond strength of 0.8. Further details in Methods.

Briefly, conformational sampling was performed on ps-$I_{Ks}$ residues D39-A44, and conformations showing the lowest free energy were selected for docking using a four-dimensional (4D) docking approach to find the best binding pose of the ligand. A conformation with the best docking score (Figure 2B) shows mefenamic acid binding to the pocket formed between extracellular KCNE1 residues, the external S6 transmembrane helix of one subunit, and the S1 transmembrane domain of the neighboring subunit and the pore turret of a third subunit. The estimated volume of the pocket in the mefenamic acid bound state is ~307 Å$^3$ with a hydrophobicity value of ~0.65 kcal/mol (Figure 2C). The free energy of the mefenamic acid-ps-$I_{Ks}$ interaction estimated by the MM/GBSA method from 300 ns MD simulations was -39.4 ± 0.3 kcal/mol (Figure 4A), while a similar value of -31.8 ± 1.46 kcal/mol was calculated using the Poisson-Boltzmann surface area (MM/PBSA) model. Mefenamic acid in this complex was stabilized by its hydrophobic and van der Waals contacts with KCNQ1 and KCNE1 subunits as well as by two hydrogen bonds formed between the drug and ps-KCNE1 residues Y46 and E43 (Figure 2D, Supplemental movie S1).

The energy decomposition per amino acid using MM/GBSA and MM/PBSA methods revealed several residues in KCNQ1 and psKCNE1 with significant contributions to mefenamic acid coordination (Figure 2D, Figure 2-figure supplement 2). As expected, these are the KCNE1 residues located at the external region of the auxiliary subunit – amino acids K41 to Y46. In addition, residues W323 and V324 located on the S6 helix as well as residues L142, Q147 and Y148 located on the S1 helix exhibited the lowest interaction free energy scores. We focused on functional validation of the KCNE1 residues K41, L42, E43, A44 and Y46, and KCNQ1 residues W323, V324, L142, Q147 and Y148 by mutation to cysteine, alanine and/or tryptophan, and examining the sensitivity of fully saturated EQ channel complexes to 100 μM mefenamic acid.

## Mutational impact on EQ current changes induced by mefenamic acid

The effect of mutations on the current waveform and tail current response to 100 μM mefenamic acid treatment was examined on $I_{Ks}$ channels identified as: x-EQ-y where "x" denotes a KCNE1 mutation and "y" denotes a KCNQ1 mutation. In the absence of mefenamic acid (control), most mutations, with the exception of EQ-L142C (Figure 3-figure supplement 1), produced slowly activating currents with rapid tail current decay (Figure 3A).

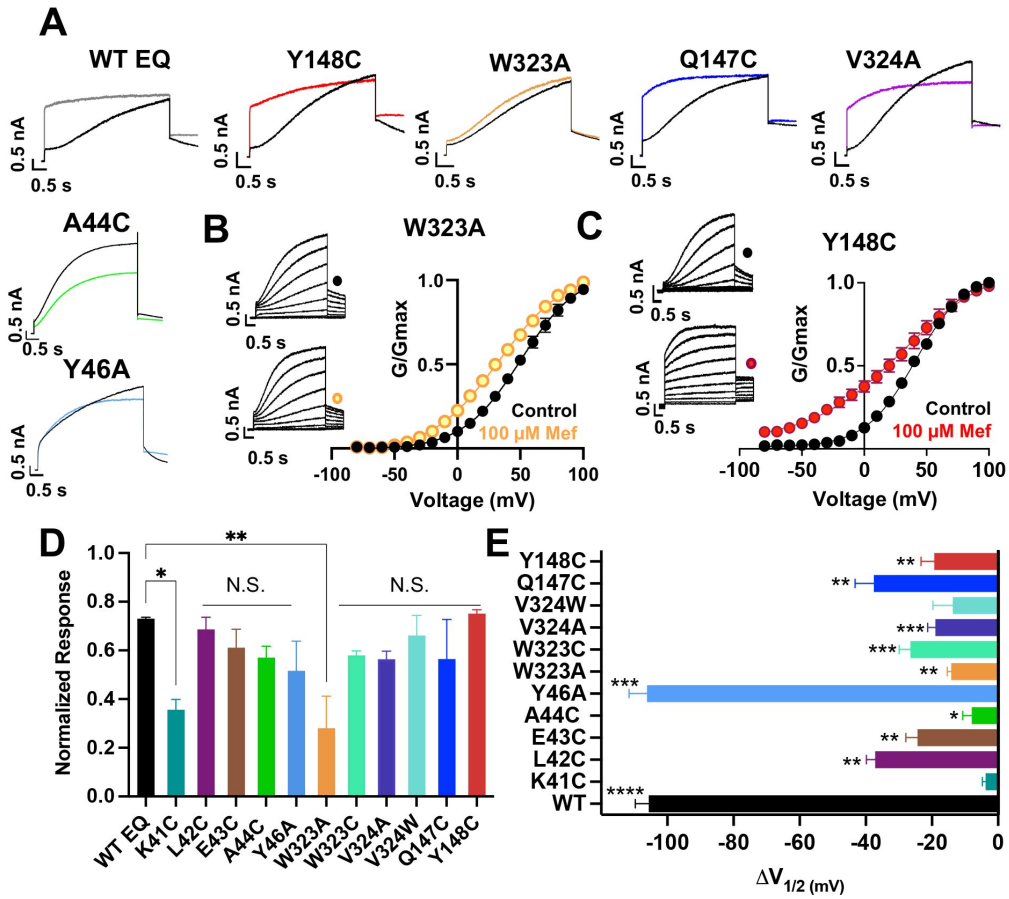

**Figure 3.**

**Current waveform and G-V changes induced by mefenamic acid in binding site mutants.**

(A) Current traces from WT EQ and key residue mutants in control (black) and 100 μM Mef (colors).

(B) EQ-W323A and (C) EQ-Y148C current traces in control (top) and presence of 100 μM Mef (below). G-V plots in control (black) and presence of 100 μM Mef (colors). Boltzmann fits were: EQ-W323A control (n = 4): $V_{1/2}$ = 47.8 mV, *k* = 23.7 mV; EQ-W323A Mef (n = 3): $V_{1/2}$ = 33.7 mV, *k* = 28.5 mV; EQ-Y148C control (n = 4): $V_{1/2}$ = 36.8 mV, *k* = 20.3 mV; and EQ-Y148C Mef (n = 4): $V_{1/2}$ = 17.5 mV, *k* = 36.7 mV). Voltage steps were from -80 mV to +100 mV for 4 s, followed by a 1s repolarization to -40 mV. Interpulse interval was 15 s.

(D) Summary plot of the normalized response to 100 μM Mef (see methods). ** and * denote a significantly reduced response compared to WT EQ. N.S. denotes not significant.

(E) Plot of $V_{1/2}$ response ($\Delta V_{1/2}$) in drug for different mutants compared with WT. Significance of differences as stated in Methods. n-values for mutants in D and E are stated in Table 1.

In the presence of 100 µM mefenamic acid, the waveforms of WT EQ, S6 and pore mutations EQ-V324A, EQ-V324W, EQ-Q147C and EQ-Y148C were all transformed into ones with instantaneous current onset and slowed tail current decay (Figure 3A). Only the EQ-W323A waveform and tail current was largely unaffected by 100 µM mefenamic acid (like K41C-EQ, Figure 1). The EQ-W323C and A44C-EQ current waveforms were also unchanged by 100 µM mefenamic acid, but their tail current decay was slowed. We interpreted this to mean that mefenamic acid binds to EQ-W323C and A44C-EQ mutant open channels and slows closing, but that the drug-channel complex is less stable and mefenamic acid unbinds during the interpulse interval, which relieves drug action between pulses. The summary of the normalized response of the different mutants to mefenamic acid is shown in Figure 3D. G-V plots were obtained from the tail current amplitudes in the absence and presence of 100 µM mefenamic acid, and unlike WT EQ (Figure 1), minimal change in the shape and position of the EQ-W323A G-V plot was seen after exposure to 100 µM mefenamic acid (V1/2 shift of -14.1 mV compared with -105.7 mV seen in WT EQ (Figure 1, Figure 3B, 3E, and Table 1). For the W323C mutation, a less drastic decrease in size compared to alanine, the V1/2 shift seen with 100 µM mefenamic acid increased (EQ-W323C ΔV1/2 = -26.5 mV), suggesting that the size of the W323 residue is important (Figure 3E and Table 1). However, when the neighboring V324 KCNQ1 residue was mutated to a smaller (alanine) or bulkier (tryptophan) residue they both showed the same response to mefenamic acid. Both the V324W and V324A mutations reduced V1/2 shifts caused by 100 μM mefenamic acid to between -14 to -19 mV (Figure 3E and Table 1).

Extracellular S1 residues identified in the MD simulations also proved to be important for mefenamic acid binding, although less so than W323 and K41 residues found in the KCNQ1 extracellular regions of the S6 segment and KCNE1, respectively. Compared to WT, EQ-Y148C reduced the V1/2 shift after exposure to 100 μM mefenamic acid (EQ-Y148C ΔV1/2 = -19.3 mV) and lessened the slope of the G-V relationship (Figure 3C). The Q147C mutant on the other hand, only partially prevented the V1/2 shift observed after mefenamic acid treatment (EQ-Q147C $\Delta V_{1/2\,=}$ -37.6 mV, Figure 3E).

Other KCNE1 residues located at the N-terminal limit of the KCNE1 transmembrane segment (L42, E43, A44, Y46) were also investigated. Unlike with K41C, but similarly to WT EQ, tail current decay was inhibited in the presence of 100 μM mefenamic acid in all of these mutants, reflected by the normalized response (Figure 3D). In addition, a reduced but statistically significant shift in $V_{1/2}$ was observed with 100 μM mefenamic acid compared to control in all mutants tested except K41C. (K41C $\Delta V_{1/2}$ = -3.8 mV; L42C $\Delta V_{1/2}$ = -37.1 mV; E43C $\Delta V_{1/2}$ = -24.4 mV; A44C $\Delta V_{1/2}$ = -9.7 mV; Figure 3D). Consistent with previous literature (Gofman et al., 2012; Wang et al., 2012; Kuenze et al., 2020), Y46C-EQ in control conditions produced a current with faster activation and a complex GV curve which made analysis and an assignment of slope and $V_{1/2}$ difficult, though a small left shift in the GV curve was visible (data not shown). In lieu of Y46C, G-V data from the Y46A-EQ mutant showed a potent effect of mefenamic acid, equivalent to that of WT with a $\Delta V_{1/2}$ = -106.2 mV (Figure 7-figure supplement 1). These results suggest that mutation of residues further away from the N-terminus of KCNE1 than K41 has diminishing effects on the activator action of mefenamic acid.

## Augmented activation of EQ-L142C in the absence of mefenamic acid

Unlike other mutations, the control EQ-L142C current waveform displayed an almost instantaneous current onset, and tail currents showed no decay with our standard protocol (Figure 3-figure supplement 1A). As the control EQ-L142C current waveform qualitatively resembled WT EQ currents in the presence of 100 μM mefenamic acid, G-V plots at different interpulse intervals were compared. At the standard interval of 15 s, the EQ-L142C G-V plot (blue) closely overlapped that of WT EQ in the presence of Mef (solid grey), and at a 30 s interpulse interval, the position of the WT EQ + Mef (grey open circles) and EQ-L142C plots (green) both depolarized significantly (Figure 3-figure supplement 1B). It appeared that the L142C mutation augmented channel activation as much as 100 μM mefenamic acid on WT channels.

In addition, mefenamic acid made the EQ-L142C G-V plot voltage independent, overlapping with the G-V obtained for EQ-L142C with a 7s interpulse interval (pink and red respectively, Figure 3-figure supplement 1B). The data indicate that EQ-L142C is still responsive to mefenamic acid but does so from a heightened state of activation (Table 2).

**Table 2.**

Mean $V_{1/2}$ of activation (mV) and slope values (*k*-factor, mV) for WT EQ treated with 100 µM mefenamic acid, and untreated mutant EQ-L142C. Interpulse interval used is as indicated. SEM is denoted by ±. An interpulse interval of 7 s created such a dramatic change in the shape of the EQ-L142C G-V plot (see Figure 3-figure supplement 1) that a Boltzmann curve could not be fit and $V_{1/2}$ and *k*-values are therefore not available.

| | $V_{1/2}$ | *k-factor* | n |
|---|---|---|---|
| **WT EQ + 100 µM mefenamic acid: Interpulse interval 15 s** | -80.3 ± 4.1 | 41.3 ± 8.4 | 3 |
| **WT EQ + 100 µM mefenamic acid: Interpulse interval 30 s** | 26.7 ± 10.5 | 66.6 ± 28 | 8 |
| **EQ-L142C control: Interpulse interval 15 s** | -80.3 ± 4.5 | 30.0 ± 2.9 | 6 |
| **EQ-L142C control: Interpulse interval 30 s** | -28.7 ± 19 | 62.5 ± 9.6 | 3 |

## MD simulations of mutant ps-$I_{Ks}$ channels exposed to mefenamic acid

To explore the role played by critical residues, *in silico* homologous mutations K41C and W323A were introduced into the mefenamic acid-bound ps-$I_{Ks}$ channel (Mef-ps-$I_{Ks}$) and the stability of the Mef-mutant complexes was assessed during 300 ns MD simulations compared to the WT-Mef complex. Remarkably, mefenamic acid detached from mutant K41C and W323A channels within 120-140 ns of MD simulations in all three independent runs (Supplemental movies S1-S3) performed for each mutant. In contrast, mefenamic acid remained bound during the entire simulation time to WT and Y46C-ps-$I_{Ks}$ complexes (Supplemental movie S4). The last mutant was tested as we could not determine the functional effect of mefenamic acid on this residue in electrophysiological experiments. Mefenamic acid unbound from A44C-ps-$I_{Ks}$ channel complexes within 80, 100 and 110 ns during three different simulations, changing binding pose several times before doing so.

A significantly *reduced* free interaction energy (ΔG) of ligand binding for all three mutant Mef-ps-$I_{Ks}$ complexes was observed compared to WT Mef-ps-$I_{Ks}$ (Figure 4A, Table 3), and the small but statistically significant change in free energy observed for the Y46C mutant complex indicates that Y46 residue is not as important as K41 and W323 for mefenamic acid binding. Furthermore, the flexibility of the external ps-KCNE1 protein residues of the mutants W323A and K41C ps-$I_{Ks}$ channels was analyzed by monitoring their average root mean square fluctuation (RMSF) during the last 100 ns of simulations after the detachment of mefenamic acid from the molecular complex. The RMSF values obtained from the two mutant channels were then compared to that of WT ps-$I_{Ks}$ channels where MD simulations of the same duration were conducted after removing the mefenamic acid molecule from the complex (Figure 4B). The results indicate that W323A and K41C mutations markedly increased the RMSF of D39-A44 ps-KCNE1 residues when compared to the WT ps-$I_{Ks}$ channel complex without mefenamic acid bound (Figure 4C, 4D). These results suggest that the side chains of K41 and W323 residues normally stabilize the conformation of the external region of KCNE1, so that mutation of these residues increases random motion and reduces the ability of drugs like mefenamic acid to remain bound at this site.

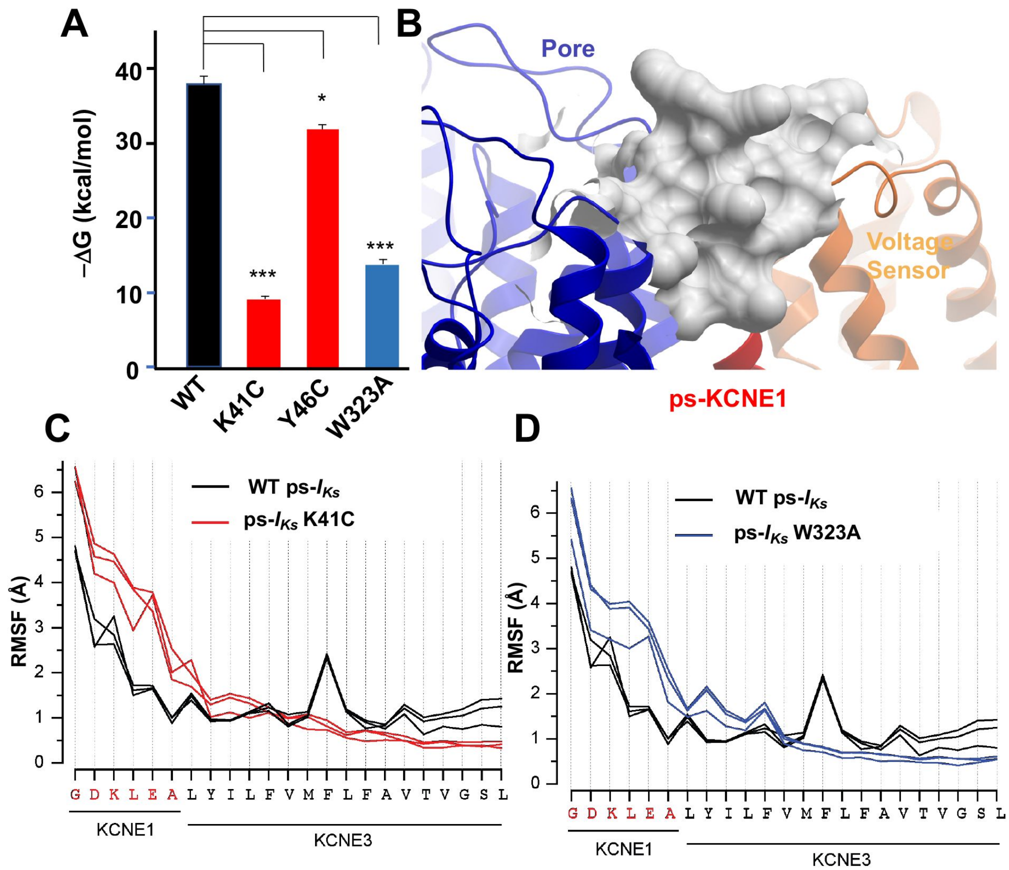


**Figure 4.**

**K41C, Y46C and W323A mutant Impact on mefenamic acid binding energy and flexibility of external KCNE1 residues.**

(A) Average free interaction energy of Mef-bound ps-$I_{Ks}$ complexes calculated according to MM/GBSA method from three independent MD simulation runs. For K41C and W323A mutations, calculations correspond to interval of simulations before the detachment of ligand from molecular complex. * and *** denote significant differences in average free interaction energy compared to WT.

(B) Surface representation of WT ps-$I_{Ks}$ after removal of Mef. The pore is blue, ps-KCNE1 is red and the VSD of a neighbouring subunit is yellow.

(C-D) Root mean square fluctuations (RMSF) of ps-KCNE residues (Å) in the ps-$I_{ks}$ complex during the last 100 ns of simulations. Three separate MD simulations shown for WT ps-$I_{Ks}$ channel without Mef (black lines) and three for K41C (C, red) and W323A (D, blue) after ligand detachment. GROMACS software was used for RMSF analysis.

**Table 3.**

Average free interaction energies of MEF-bound ps-$I_{Ks}$ complexes calculated according to MM/GBSA methods from three independent MD simulation runs. Mean values are in kcal/mol, ± SD. For W323A and K41C mutations, calculations correspond to interval of simulations before the detachment of ligand from molecular complex. Note that unbinding occurred in K41C and W323A in all 3 simulations, e.g. Supplemental movies S2 and S3.

| Run | Method | Wild Type | W323A | Y46C | K41C |
|---|---|---|---|---|---|
| I | GBSA | -39.3 ± 4.1 | -13.0 ± 5.1 | -31.2 ± 2.9 | -10.5 ± 3.8 |
| | | - | Unbinding after ~75 ns | - | Unbinding after ~25 ns |
| II | GBSA | -39.0 ± 3.3 | -23.1 ± 3.1 | -32.2 ± 3.8 | -17.0 ± 2.6 |
| | | - | Unbinding after ~120 ns | - | Unbinding after ~70 ns |
| III | GBSA | -35.5 ± 2.6 | -22.6 ± 4.1 | -28.8 ± 5.7 | -17.2 ± 2.5 |
| | | - | Unbinding after ~85 ns | - | Unbinding after ~20 ns |

## Mutational impact on $I_{Ks}$ current changes induced by DIDS

To establish whether the binding pocket for mefenamic acid can be generalized to other $I_{Ks}$ activators, we also examined the binding of the structurally unrelated $I_{Ks}$ activator, DIDS. Stilbenes such as DIDS (Figure 5A) also activate $I_{Ks}$ (Abitbol et al., 1999; Bollmann et al., 2020), and given their molecular differences from fenamates, it is of interest to explore common structural and dynamic features of their binding to the $I_{Ks}$ channel complex. 100 µM DIDS had no effect on endogenous currents in GFP-transfected tsA201 cells but treatment of WT EQ channels with 100 µM DIDS transformed the slowly activating waveform into one with faster onset (although not instantaneous like mefenamic acid) and inhibited tail current decay (Figure 5B). G-V plots were obtained from the tail amplitudes in the absence and presence of 100 µM DIDS (Figure 5C, 5D). The overall shape, slope, and $V_{1/2}$ of the WT EQ G-V relationship changed with 100 µM DIDS in the direction of increased activation ($\Delta V_{1/2}$ = -46.6 mV; Table 4; control $k$ = 20.3 mV, DIDS $k$ = mV). However, the effects on WT EQ were less pronounced with DIDS than mefenamic acid at the 100 µM concentration.

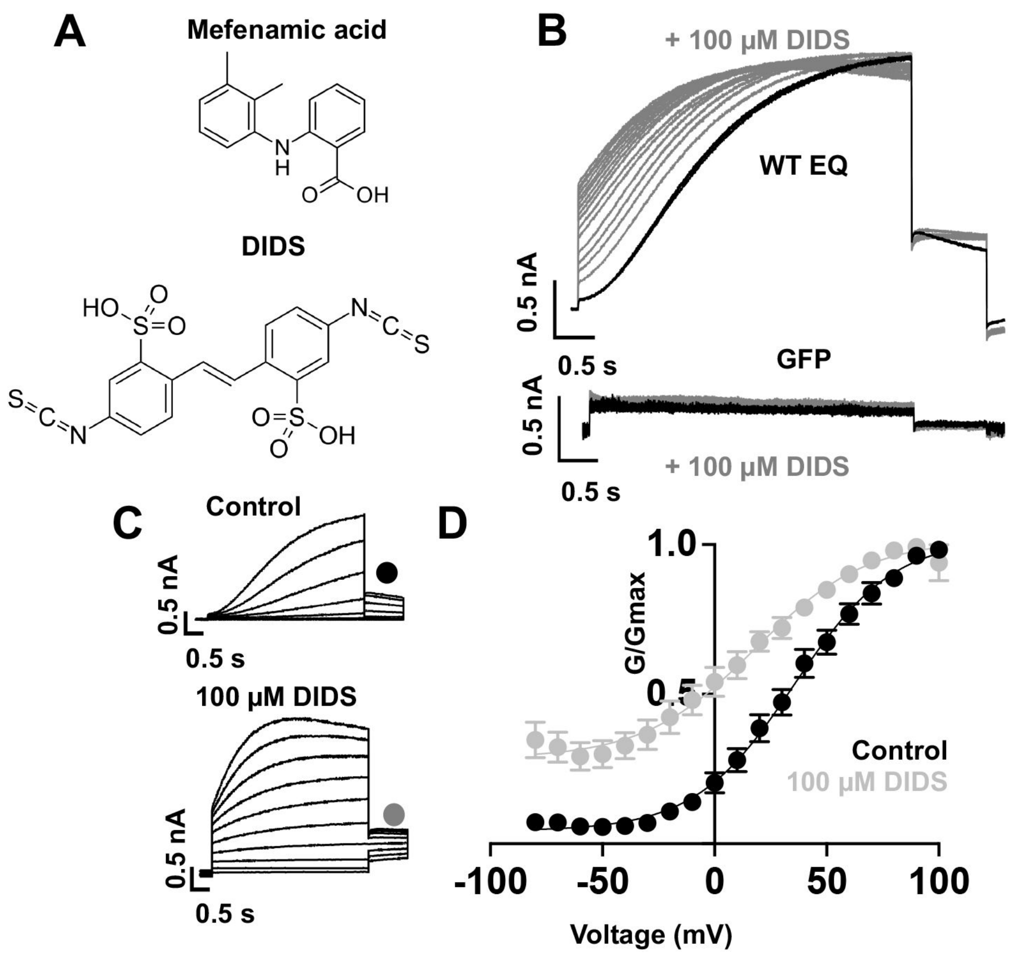


**Figure 5.**

**Effect of DIDS on $I_{Ks}$.**

(A) Molecular structure of mefenamic acid and DIDS.

(B) WT EQ current in control (black) and exposed to 100 µM DIDS over time (grey). Pulses were from -80 to +60 mV every 15 s, and current traces are shown superimposed. Lower panel shows no effect on currents from GFP-only transfected cell exposed to 100 µM DIDS over time (grey).

(C) Current traces from WT EQ in control and presence of 100 µM DIDS as indicated. Pulses were from -80 to +100 mV for 4s, with a 1 s repolarization to -40 mV. Interpulse interval was 15 s.

(D) Corresponding G-V plot in control (black) and DIDS (grey) from data as shown in panel C. Boltzmann fits were: WT EQ control (n = 8): $V_{1/2}$ = 30.5 mV, $k$ = 20.3 mV; WT EQ in the presence of DIDS (n = 5): $V_{1/2}$ = -16.1 mV, $k$ = 25.3 mV.

**Table 4.**

Mean $V_{1/2}$ of activation (mV) and slope values (*k*-factor, mV) in the absence and presence of 100 µM DIDS for fully saturated $I_{Ks}$ channel complexes. A statistical difference in $V_{1/2}$ compared to control is shown as p-value determined using an unpaired t-test. NS denotes not significant. SEM is denoted by ±

| | Control | | | 100 µM DIDS | | | p-value |
|---|---|---|---|---|---|---|---|
| | $V_{1/2}$ | *k-factor* | n | $V_{1/2}$ | *k-factor* | n | |
| **Wildtype EQ** | 30.5 ± 4.3 | 20.3 ± 0.9 | 8 | -16.1 ± 2.8 | 25.3 ± 1.9 | 5 | <0.001 |
| **K41C-EQ** | 23.1 ± 3.0 | 20.2 ± 1.8 | 3 | -1.6 ± 3.7 | 24.0 ± 3.6 | 3 | <0.01 |
| **L42C-EQ** | 55.4 ± 3.3 | 19.9 ± 1.5 | 3 | 21.4 ±10.7 | 28.5 ± 2.6 | 4 | <0.05 |
| **A44C-EQ** | 24.1 ± 1.7 | 29.6 ± 3.3 | 4 | 5.6 ± 2.4 | 17.6 ± 1.8 | 3 | <0.05 |
| ***Y46A-EQ** | 75.2 ± 2.1 | 59.1 ± 5.3 | 5 | 25.8 ± 0.9 | 39.2 ± 1.4 | 5 | <0.0001 |
| **EQ-Y148C** | 51.8±4.6 | 23.5±1.7 | 4 | 40.0 ± 3.1 | 26.3 ± 1.5 | 5 | **NS** |
| **EQ-W323A** | 50.7 ± 3.6 | 23.9 ± 1.6 | 4 | 23.4 ± 4.9 | 24.1 ± 1.3 | 4 | <0.05 |

* An estimation of the activation $V_{1/2}$ was calculated from a right shifted, non-saturating GV curve.

The ps-$I_{Ks}$ construct and docking procedures used previously to explore the binding site of mefenamic acid were used as a basis for DIDS docking and subsequent MD simulations. We found that DIDS also bound to a location formed between extracellular KCNE1 residues, the external S6 transmembrane helix of one subunit, the S1 transmembrane domain of the neighboring subunit, and the pore turret of the opposite subunit (Figure 6A). DIDS is stabilized by its hydrophobic and van der Waals contacts with KCNQ1 and KCNE1 subunits as well as by two hydrogen bonds formed between the drug and ps-KCNE1 residue L42 and KCNQ1 residue Q147. The energy decomposition per amino acid using MM/GBSA and MM/PBSA methods revealed several residues in KCNQ1 and psKCNE1 with significant contributions to DIDS coordination (Figure 6B). Although residues identified as critical to mefenamic acid binding and action also generally appear to be important for DIDS, especially W323, K41 was noticeably not as critical to DIDS coordination whereas A44 and Y46 were more so.

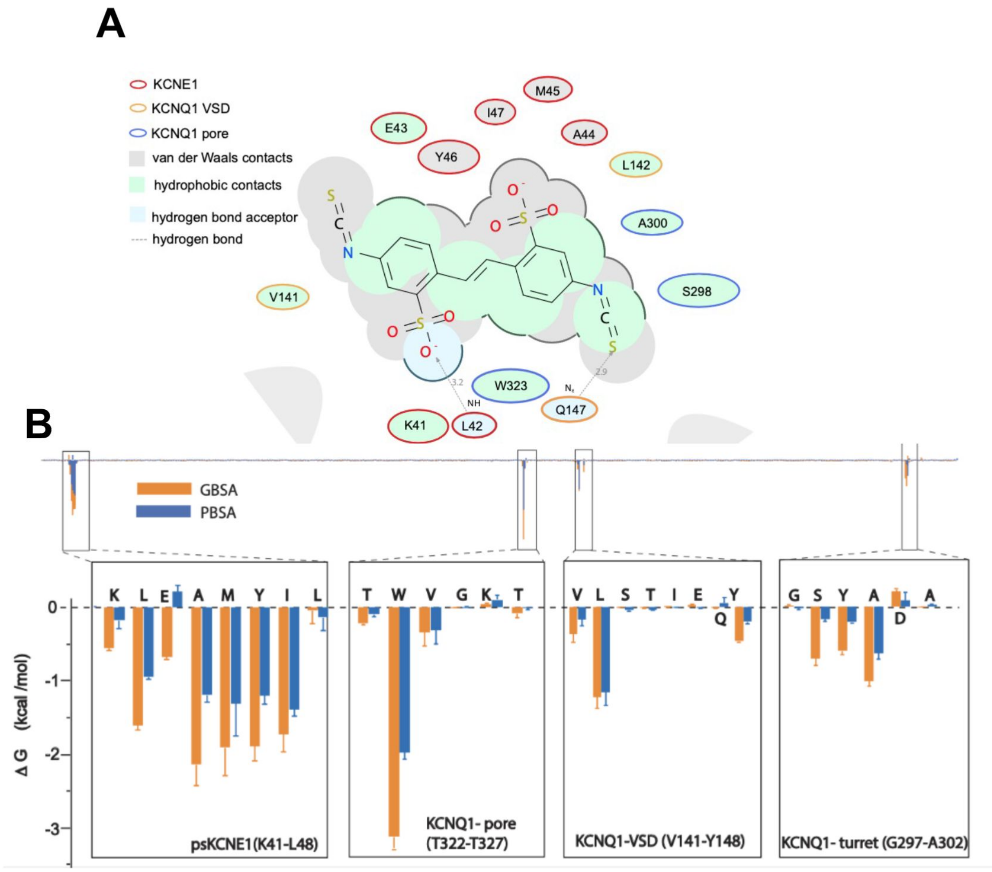


**Figure 6.**

**Ligand interaction and energy decomposition per amino acid for DIDS binding to ps-$I_{Ks}$.**

(A) Ligand interaction map of DIDS with ps-$I_{Ks}$. Size of residue ellipse is proportional to the strength of the contact. The distance between the residue label and ligand represents proximity. Grey parabolas represent accessible surface for large areas. The 2D diagram was generated by ICM pro software with a cut-off value for hydrophobic contacts 4.5 Å and hydrogen bond strength 0.8.

(B) Energy decomposition per amino acid for DIDS binding to ps-$I_{Ks}$. Generalized Born Surface Area (GBSA; orange) and Poisson-Boltzmann Surface Area (PBSA; blue) methods were used to estimate the interaction free energy contribution of each residue in the DIDS-bound ps-$I_{Ks}$ complex. The lowest interaction free energy for residues in ps-KCNE1 and selected KCNQ1 domains are shown as enlarged panels (n=3 for each point).

Similar to mefenamic acid, DIDS stayed bound to WT ps-$I_{Ks}$, but detached from the W323A mutant during two of three runs, and from Y46C channels in one of three runs after 240 and 122 ns of simulation, respectively (Supplemental movies S5, S7). Interestingly, DIDS did not unbind from K41C mutant ps-$I_{Ks}$ channels, even after 250 ns of simulation, although the binding pose did change in one run (Supplemental movie S6). Calculations of free interaction energy shown in Table 5 using both PBSA and GBSA methods are of mean values before the detachment of the drug if that occurred. If DIDS detachment did not occur during 250 ns MD runs, average free energy was calculated over the duration of the entire MD run for that construct. Only DIDS binding to W323A, and to Y46C in one simulation run, showed significantly lower free energy than DIDS binding to WT, using both methodologies. The small change in free energy and more stable binding of DIDS to K41C suggest that this residue is not as important as W323 or Y46 for DIDS binding.

**Table 5.**

Average free interaction energies of DIDS-bound ps-$I_{Ks}$ complexes calculated according to MM/PBSA and MM/GBSA methods from three independent MD simulation runs. Mean values are in kcal/mol, ± SD. For W323A and Y46C mutations, calculations correspond to interval of simulations before the detachment of ligand from molecular complex. Note that unbinding occurred in W323A in 2 of 3 simulations, and K41C did not unbind but the binding pose shifted – see Supplemental movie S6.

| Run | Method | Wild Type | W323A | Y46C | K41C | A44C | Y148C |
|---|---|---|---|---|---|---|---|
| I | PBSA | -32.3 ± 6.7 | -25.7 ± 4.9 | -35.5 ± 6.3 | -33.5 ± 4.4 | -29.2 ± 4.2 | -34.1± 4.6 |
| | GBSA | -35.5 ± 4.6 | -30.2 ± 4.5<br>unbinding after 240 ns | -42.7 ± 5.4 | -41.5 ± 5.8<br>changed binding pose after ~140-150 ns | -36.8 ± 2.9 | -40.7 ± 4.0 |
| II | PBSA | -32.6 ± 4.8 | -25.3 ± 5.5 | -25.7 ± 6.5 | -36.4 ± 4.2 | -29.0 ± 4.0 | -29.3 ± 3.7 |
| | GBSA | -36.0 ± 4.8 | -30.0± 4.8 | -32.4 ± 6.4<br>unbinding after ~100-132 ns | -31.6 ± 4.0 | -34.0 ± 5.7 | -31.8 ± 4.7 |
| III | PBSA | -29.2 ± 8.1 | -27.1± 3.9 | -38.3 ± 4.8 | -35.1 ± 4.3 | -31.2 ± 45.0 | -35.7 ± 5.8 |
| | GBSA | -34.0 ± 8.9 | -35.1± 4.8<br>unbinding after ~240 ns | -42.5 ± 4.7 | -30.8 ± 3.9 | -35.6 ± 5.6 | -40.3 ± 5.9 |

These model predictions were tested experimentally using fully saturated mutant $I_{Ks}$ channel complexes, K41C-EQ, L42C-EQ, A44C-EQ, Y46A-EQ, EQ-W323A, EQ-L142C and EQ-Y148C (e.g. Figure 7A). Y46 was identified as an important residue for DIDS binding in MD simulations, but as before, the complexity of the GV curve and low functional expression of the Y46C construct led us to substitute Y46A for this mutation in electrophysiology experiments (Figure 7 and Figure 7-figure supplement 1).

K41C-EQ G-V plots were obtained from peak tail current amplitudes in the absence and presence (Figure 7B) of 100 μM DIDS. Unlike with mefenamic acid, but in agreement with the modeling, K41C did not prevent the action of DIDS. Treatment of K41C-EQ with 100 μM DIDS hyperpolarized the V1/2 and changed the shape of the G-V relationship compared to control (Figure 7B, 7D, Table 4). The current waveform of K41C-EQ activated more quickly with less sigmoidicity after treatment with DIDS, and tail current decay was slowed as indicated by the normalized response (Figure 7B, 7C). As K41C-EQ remained responsive to DIDS, and L42C responded similarly to K41C-EQ (for K41C-EQ, $\Delta V_{1/2}$ was -24.7 mV, and for L42C-EQ $\Delta V_{1/2}$ was -34 mV), we hypothesized that DIDS binding to $I_{Ks}$ channel complexes involved KCNE1 residues closer to the transmembrane region as predicted by the ligand interaction map for DIDS (Figure 6A). In agreement with this idea, A44C showed little response to 100 μM DIDS both in term of current waveform and tail current decay. The shift in $V_{1/2}$ was also reduced compared to WT-EQ (A44C-EQ $\Delta V_{1/2}$ = -18.5 mV; WT EQ $\Delta V_{1/2}$ = -46.6mV, Figure 7C, D). In Y46A, the effect of DIDS on the current waveform was greatly reduced as was slowing of tail current decay (Figure 7A, 7C). We could only estimate an activation $V_{1/2}$ for Y46A as the G-V curve was right shifted to very positive potentials and non-saturating. Nevertheless, hyperpolarization of the $V_{1/2}$ (Figure 7D, Figure 7-figure supplement 1) in the presence of DIDS was observed, suggesting that unstable and short-lived binding of the drug to the channel complex was sufficient to interfere with channel gating.

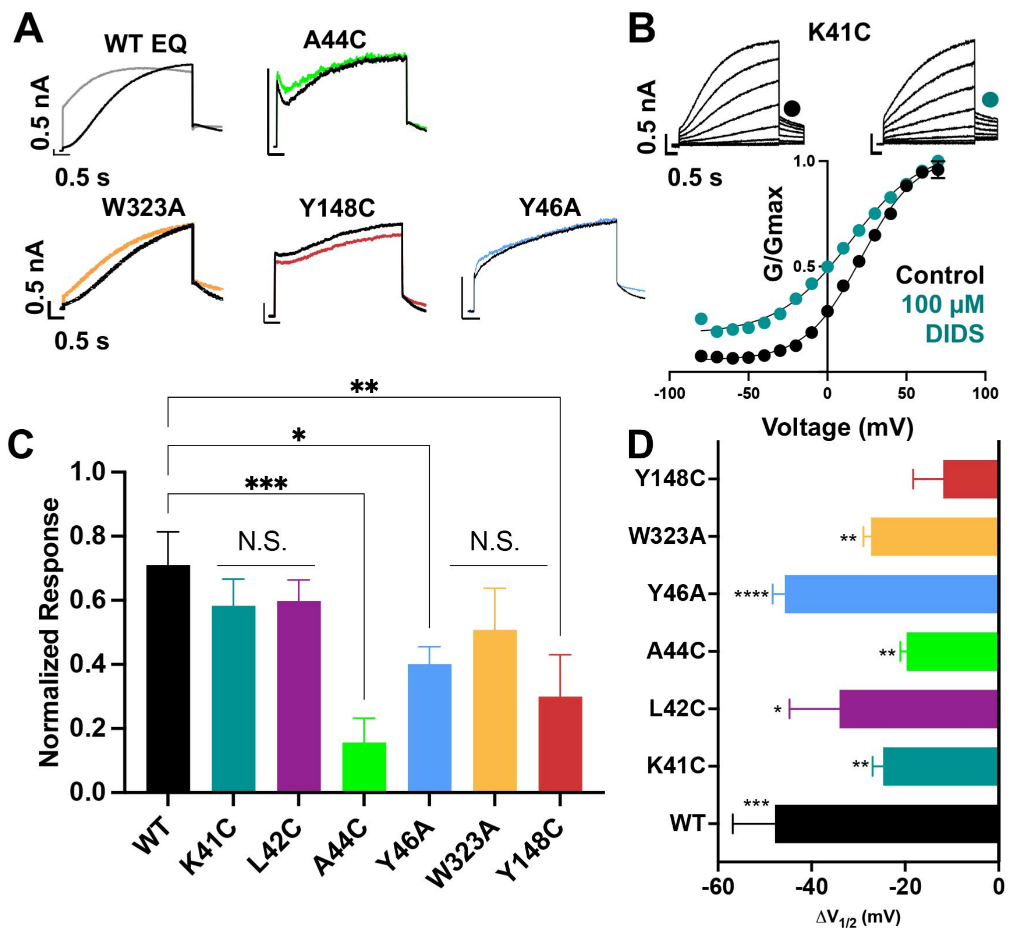


**Figure 7.**

**Binding site mutants important for the action of DIDS.**

(A) Current traces from WT EQ and key mutants in the absence (control; black) and presence (colors) of 100 µM DIDS. All calibration bars denote 0.5 nA and 0.5 s.

(B) Data and G-V plots in control (black) and DIDS (teal). Boltzmann fits were: K41C-EQ control (n = 3): $V_{1/2}$ = 23.1 mV, $k$ = 20.2 mV; K41C-EQ DIDS (n = 5): $V_{1/2}$ = -1.6 mV, $k$ = 24.0 mV. Voltage steps from a holding potential of -80 mV to +70 mV for 4 s, followed by repolarization to -40 mV for 1 s. Interpulse interval was 15 s. All calibration bars denote 0.5 nA and 0.5 s.

(C) Summary plot of the normalized response to 100 µM DIDS. For calculation, see Methods.

(D) $V_{1/2}$ response ($\Delta V_{1/2}$) to 100 µM DIDS in different mutants compared to WT. Voltage protocol was as in panel B. n-values for mutants in C and D are stated in Table 4.

The modeling studies (Figure 6) suggested that W323 remained a key residue for $I_{Ks}$ activator sensitivity, and in agreement with this, EQ-W323A was only partially responsive to DIDS. In EQ-W323A, the tail current decay was affected but the slowly-activating current waveform was preserved in the presence of DIDS, supporting the idea that the drug dissociates from the channel complex between pulses (Figure 7A, 7C, Table 5). However, a significant shift in the V1/2 remained ($\Delta V_{1/2}$ = -27.3 mV; Figure 7D, Table 4). Comparable effects observed in EQ-W323A, A44C-EQ, and Y46A-EQ are consistent with the idea that the DIDS binding pocket involves the N-terminal end of S6 and the deeper extracellular surface of the KCNE1 transmembrane segment. In EQ-Y148C, the $V_{1/2}$ and the shape of the G-V plot were not altered by 100 µM DIDS (Figure 7D, Table 4), and tail current decay was not markedly slowed

(Figure 7A, 7C). This result suggests that unlike mefenamic acid, DIDS binding determinants extend to the S1 region of KCNQ1 and that Y148 is an important contributor.

Notably, as with mefenamic acid, the L142C-EQ G-V plot became voltage-independent after treatment with 100 μM DIDS (data not shown) and so it proved hyper-responsive to both of the $I_{Ks}$ activators, DIDS and mefenamic acid.

# Discussion

Mefenamic acid, a nonsteroidal anti-inflammatory drug (NSAID), and the structurally distinct stilbenes DIDS and SITS, among other compounds, have previously been identified by numerous groups to enhance $I_{Ks}$ currents in various expression systems including canine and guinea-pig ventricular myocytes (Magyar et al., 2006; Toyoda et al., 2006) as well as heterologous expression systems such as *Xenopus laevis* oocytes, CHO, COS-7, tsA201 and LM cells (Busch et al., 1994; Abitbol et al., 1999; Unsöld et al., 2000; Toyoda et al., 2006; Wang et al., 2020). The extracellular region of KCNE1 between residues 39-44 was found important in mediating the effect of DIDS on $I_{Ks}$ (Abitbol et al., 1999) and residue K41, located on the extracellular end of KCNE1, was found to be critical in mediating mefenamic acid's activating effect on the fully saturated $I_{Ks}$ channel complex (Wang et al., 2020). Consistent with this idea, when all four WT KCNE1 subunits are replaced with mutant K41C-KCNE1, mefenamic acid up to a concentration of 1 mM is largely ineffective (Figure 1). Unlike in the WT complex, there is little change to the current waveform, slope or $V_{1/2}$ of the G-V plot during activator exposure, suggesting that all the drug binding site(s) on the channel complex are impaired, or that the mechanism of action is disabled, or that the mutation causes a combination of these actions. In the present study, we analysed these activator compound actions using molecular modeling approaches, and complementary mutational analysis. Our data describe the formation of a drug binding pocket between the immediate extracellular residues of KCNE1, S1 and pore residues of two KCNQ1 subunits, stabilized by the presence of structurally different activator compounds. The effect of mutations that negate the activator compound actions is to destabilize the binding pocket itself, reduce drug binding and limit activator residency time on the channel complex.

## Mefenamic acid binding site in $I_{Ks}$

Using data that suggested an extracellular binding site for mefenamic acid was at the KCNE1-channel interface, MD simulations were used to explore drug-channel interactions further. The drug binding pocket was defined as the extracellular space formed by KCNE1, the domain-swapped S1 helices of one KCNQ1 subunit and the pore/turret region made up of two other KCNQ1 subunits (Figure 2; Supplemental movie S1). Docking and molecular simulations identified a binding site made up of several residues across all of these elements (Figure 2 and Figure 2-figure supplement 2, Table 6). Mutagenesis revealed that most mutations impacted at least one out of the three mefenamic acid effects: V1/2 shift, GV-plot slope change, and current waveform change related to slowed deactivation. However, only three of the mutants impacted all of the mefenamic acid actions (Figure 3). These were W323A in KCNQ1, as well as K41C, and A44C in KCNE1, which define key components of the binding pocket for mefenamic acid. That W323 forms an important medial wall of the hydrophobic pocket to which mefenamic acid binds is suggested by data from mutants in which the size of the side chain at this location is systematically made smaller to cysteine then alanine, which results in diminished effectiveness of the drug, particularly with alanine. This mutation resulted in the detachment of mefenamic acid as well as lower interaction energy of drug-channel complexes during MD simulations (Figure 4A; Supplemental movie S3) particularly loss of interaction strength with KCNE1 (Table 6). Likewise, MD simulations revealed detachment of mefenamic acid and reduced interaction energy values for K41C-Mef-$I_{ks}$ complexes (Supplemental movie S2, Figure 4A) again accompanied by loss of interaction strength with KCNE1 (Table 6). The drug also detached from the A44C complex, another mutant with a reduced response in terms of changes in waveform, $V_{1/2}$ and slope. In contrast, while the free interaction energy of the mutant Y46C channel was significantly decreased in comparison to the WT channel complex (Figure 4A), detachment of mefenamic acid from the mutant channel during 300 ns MD simulations was not observed, though S1 appears to pull away from the pore (Supplemental movie S4). Comparison of energies at specific residues show a loss of interaction with Q147 and Y148 in S1 with this mutant and a small shift towards the turret at S298 and A300 (Figure 8A; Table 6). While we were not able to obtain good electrophysiological data from Y46C, data from Y46A suggested that mefenamic acid still had potent actions on this mutant. The combination of binding studies and electrophysiological data indicate that Y46 was not as important for mefenamic acid binding and action as other KCNE1 N-terminal residues.

Table 6:

## Average free energy of mefenamic acid and DIDS bound WT and mutant ps-$I_{Ks}$ complexes.

Values are calculated according to the MM/GBSA method from three independent MD simulation runs and further broken down by residue and channel region. For K41C and W323C, calculations correspond to the interval of simulations before detachment of the ligand from the complex. Values are in kcal/mol. Mutated residues are in bold italics.

| | Residue | WT MEF | K41C MEF | Y46C MEF | W232A MEF | WT DIDS | A44C DIDS | Y46C DIDS | Y148C DIDS | W323A DIDS |
|---|---|---|---|---|---|---|---|---|---|---|
| **Total** | | -38.02 | -16.39 | -31.40 | -20.31 | -34.29 | -35.40 | -39.16 | -37.60 | -34.33 |
| **Drug** | | -19.69 | -8.79 | -15.25 | -11.65 | -17.42 | -17.52 | -21.21 | -18.18 | -19.17 |
| **Channel** | | -17.91 | -6.52 | -16.61 | -7.55 | -19.21 | -18.40 | -16.94 | -19.21 | -15.10 |
| **Ch+Drug** | | -37.60 | -15.31 | -31.86 | -19.20 | -36.63 | -35.92 | -38.15 | -37.39 | -34.27 |
| **Difference** | | -0.42 | -1.08 | 0.46 | -1.11 | 2.34 | 0.52 | -1.01 | -0.21 | -0.06 |
| E1/E3 | **K41** | -1.548 | ***-0.349*** | -1.411 | -0.954 | -0.639 | -0.439 | -1.148 | -0.283 | 0.083 |
| | **L42** | -0.970 | -0.895 | -0.943 | -1.193 | -1.061 | -0.798 | -0.165 | -0.320 | -0.587 |
| | **E43** | -2.681 | -0.472 | -2.257 | -0.309 | -1.312 | -0.660 | -0.014 | -2.264 | -0.705 |
| | **A44** | -1.906 | -0.886 | -2.035 | -1.102 | -1.778 | ***-2.062*** | -1.027 | -1.061 | -1.780 |
| | **M45** | -1.744 | -0.163 | -1.913 | -0.064 | -1.756 | -2.154 | -1.222 | -2.810 | -1.633 |
| | **Y46** | -2.391 | -0.121 | ***-2.702*** | -0.083 | -1.985 | -1.797 | ***-1.245*** | -2.377 | -1.603 |
| | **I47** | -0.771 | -0.136 | -0.412 | -0.203 | -2.036 | -1.993 | -1.004 | -0.621 | -1.528 |
| | **L48** | -0.010 | -0.003 | -0.025 | 0.005 | -0.045 | -0.100 | -0.233 | -0.081 | -0.035 |
| **Sum** | | **-12.02** | **-3.03** | **-11.70** | **-3.90** | **-10.61** | **-10.00** | **-6.06** | **-9.82** | **-7.79** |
| Pore | **T322** | -0.125 | -0.047 | -0.091 | -0.968 | -0.200 | -0.213 | -0.127 | -0.152 | -0.410 |
| | **W323** | -2.159 | -0.736 | -2.226 | ***-0.932*** | -3.312 | -3.395 | -2.279 | -3.566 | ***-1.590*** |
| | **V324** | -0.644 | -0.037 | -0.123 | -0.047 | -0.172 | -0.210 | -0.111 | -2.016 | -0.063 |
| | **G325** | -0.038 | -0.006 | -0.006 | 0.017 | -0.004 | 0.000 | 0.003 | -0.048 | 0.032 |
| | **K326** | -0.185 | -0.015 | -0.024 | 0.008 | 0.038 | 0.058 | 0.091 | 0.000 | 0.118 |
| | **T327** | -0.130 | -0.015 | -0.020 | 0.027 | -0.040 | -0.022 | -0.018 | -1.298 | -0.014 |
| **Sum** | | **-3.28** | **-0.86** | **-2.49** | **-1.89** | **-3.69** | **-3.78** | **-2.44** | **-7.08** | **-1.93** |
| S1 | **V141** | -0.061 | -0.014 | -0.247 | -0.006 | -0.569 | -0.411 | -0.660 | -0.076 | -0.504 |
| | **L142** | -0.861 | -0.434 | -1.052 | -0.267 | -1.525 | -0.822 | -1.285 | -0.460 | -1.036 |
| | **S143** | -0.028 | 0.004 | -0.014 | -0.063 | -0.008 | -0.006 | -0.104 | 0.026 | -0.017 |
| | **T144** | -0.091 | -0.006 | -0.242 | -0.526 | -0.015 | -0.122 | -0.982 | -0.002 | -0.001 |
| | **I145** | 0.004 | 0.005 | 0.002 | 0.009 | 0.003 | -0.329 | -0.067 | -0.009 | -0.001 |
| | **E146** | 0.000 | 0.048 | 0.056 | 0.114 | 0.028 | -0.014 | -1.601 | 0.113 | 0.013 |
| | **Q147** | -0.230 | -0.365 | 0.006 | -0.244 | 0.012 | -0.143 | -1.061 | -0.974 | -0.054 |
| | **Y148** | -0.565 | -1.556 | 0.006 | -0.128 | -0.428 | -0.336 | 0.002 | ***-0.022*** | -0.223 |
| **Sum** | | **-1.83** | **-2.32** | **-1.49** | **-1.11** | **-2.50** | **-2.18** | **-5.76** | **-1.40** | **-1.82** |
| Turret | **G297** | -0.024 | -0.001 | -0.010 | -0.027 | 0.016 | -0.027 | -0.089 | -0.001 | -0.076 |
| | **S298** | -0.320 | -0.163 | -0.419 | -0.427 | -0.791 | -0.848 | -1.107 | -0.216 | -1.026 |
| | **Y299** | -0.103 | -0.046 | -0.104 | -0.052 | -0.718 | -0.598 | -0.563 | -0.167 | -0.673 |
| | **A300** | -0.618 | -0.183 | -0.636 | -0.431 | -1.130 | -1.112 | -0.751 | -0.652 | -1.093 |
| | **D301** | 0.311 | 0.075 | 0.248 | 0.299 | 0.231 | 0.172 | -0.181 | 0.136 | -0.683 |
| | **A302** | -0.024 | -0.003 | -0.011 | -0.007 | -0.011 | -0.020 | 0.009 | -0.008 | -0.015 |
| **Sum** | | **-0.78** | **-0.32** | **-0.93** | **-0.65** | **-2.40** | **-2.43** | **-2.68** | **-0.91** | **-3.57** |

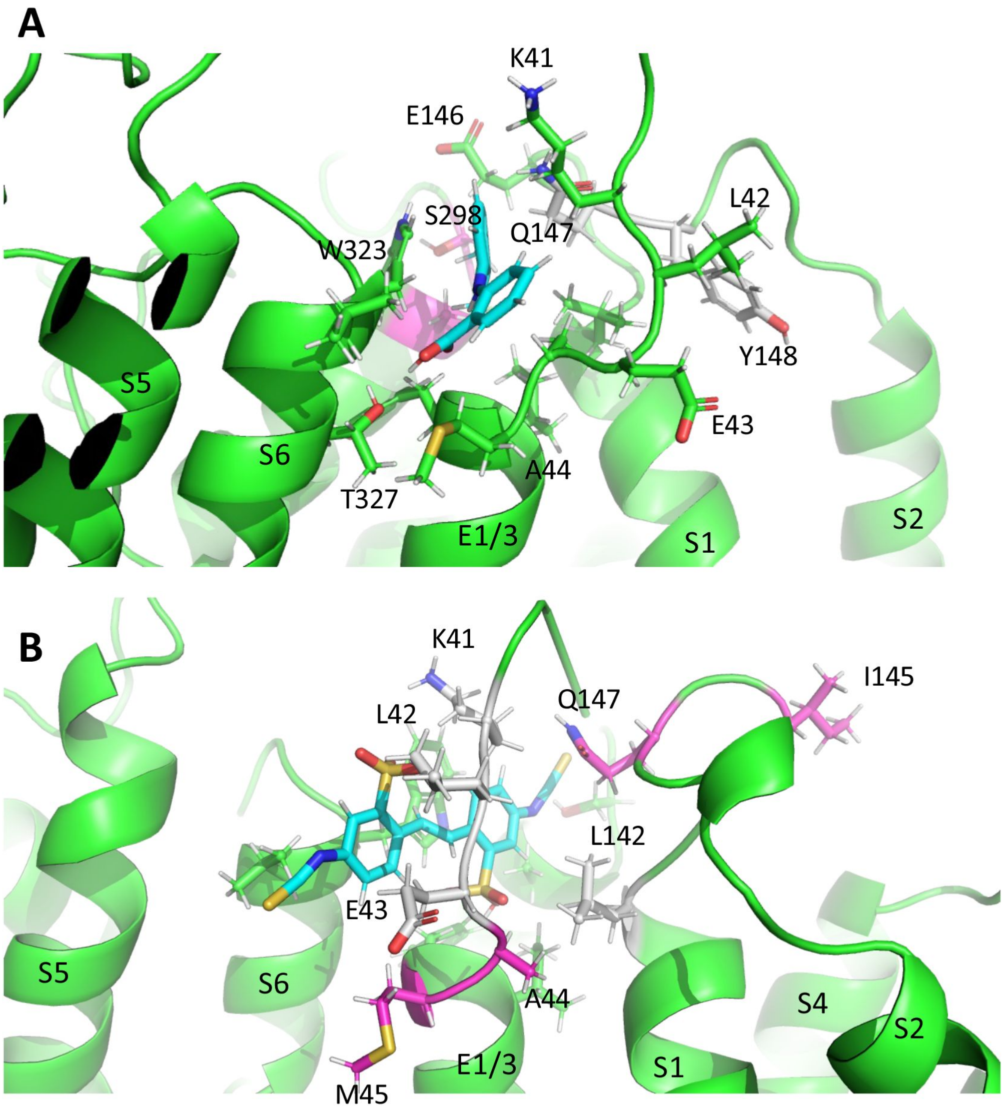


**Figure 8.**

**Subtle differences in activator binding to Y46C- and A44C-ps-IKs mutant channels.**

(A) Mefenamic acid bound to the WT ps-IKs channel. Residues that are part of the binding site are shown in stick format colored green, except those residues that were important in the WT channel that had reduced ΔG in Y46C (in grey). Residues that had slight increases in ΔG values in Y46C are shown in magenta (S298, A300). Mefenamic acid is in cyan.

(B) DIDS bound to the WT ps-IKs channel. Residues that are part of the binding site are shown in stick format colored green except those residues that were important in the WT channel that had reduced ΔG in the A44C mutant (in grey). Residues that had slight increases in ΔG values in A44C are shown in magenta. DIDS is in cyan. Images were made with the PyMOL Molecular Graphics System, Version 2.0 Schrödinger, LLC.

## Common binding site for $I_{Ks}$ activators, mefenamic acid and DIDS

Although it was suggested that DIDS and mefenamic acid have the same binding site (Abitbol et al., 1999), the extent of overlap was unknown. Initially, we examined the effect of DIDS on WT EQ. Consistent with previous studies (Abitbol et al., 1999; Bollmann et al., 2020) we confirmed that 100 µM DIDS enhanced WT EQ activity (Figure 5B) with a V1/2 shift of -46.6 mV and a decrease in the slope of the G-V curve (Table 4). As most $I_{Ks}$ activators are dependent on the KCNQ1-KCNE1 stoichiometry, this more potent effect of DIDS seen in our study may be explained by the higher dose we used and our KCNE1-KCNQ1 linked channel constructs, which ensured fully KCNE1-saturated complexes. Furthermore, due to the large size and complex folds found on the surface of oocytes, higher drug concentrations than those used for cultured cells are often required in order to facilitate a similar effect in both expression systems (Kvist et al., 2011). All previous studies utilized *Xenopus laevis* oocytes, whereas in this study transiently transfected tsA201 cells were used.

*In silico* DIDS docking revealed that both mefenamic acid and DIDS bind to the same general extracellular inter-subunit interface with some differences in key residues revealed by the electrophysiology data. Mutation Y148 in KCNQ1 to a cysteine (Y148C) or Y46 in KCNE1 to an alanine (Y46A) was found to reduce effects of 100 μM DIDS, particularly in the case of Y148C (Figure 7). These data suggest that DIDS resides deeper in the binding pocket formed by KCNE1 residues and the KCNQ1 pore/S6, and so was less dependent on the side chain of K41 to retain the activator on the channel complex (compare Movies S1 and S6). DIDS associated more strongly with the pore of the Y148C mutant, particularly with V324 and T327, and less across the KCNE1/3, S1 and turret regions (Table 6). The Y46C mutation resulted in shifts away from KCNE1/3 and towards the S1 domain (T144, E146 and Q147). In the case of A44C and DIDS, the changes were more subtle, with stronger interactions moving from central residues in KCNE1 (K41, L42 and E43) and S1 (V141 and L142; coloured grey in Figure 8B) to more peripheral residues (I145, E146 and Q147) and lower in the KCNE TMD (A44 and M45; coloured magenta in Figure 8B).

## Proposed mechanism of action for mefenamic acid and DIDS

Molecular modeling and docking revealed that mefenamic acid and DIDS induce conformational changes in the channel upon binding, to shape a binding pocket formed by residues from the external S1 domain, KCNE1 and the pore domain of $I_{Ks}$ (Figure 2C, Supplemental movies S1, S5). This cryptic binding pocket is not detectable in the absence of the drug (Figure 4B), which suggests that it has been induced in a similar manner to previous reports of toxin interactions with KcsA-Kv1.3 that induce conformational changes in both the toxin and the channel structure to generate a high-affinity binding site (Lange et al., 2006; Zachariae et al., 2008). Given the high modeled binding energy of the channel-drug activated state complexes, ~-39 kcal/mol for mefenamic acid and ~-35 kcal/mol for DIDS, but the relatively low affinities, given the micromolar concentrations required experimentally, we can imagine a dynamic complex where mefenamic acid and DIDS bind/unbind from $I_{Ks}$ at high frequency. Slowed deactivation may therefore be the result of these rapid binding/unbinding interactions slowing the dissociation of the S1/KCNE1/pore domain/drug complex by either providing steric hindrance to dissociation or by stabilizing the activated complex. MD simulations suggest the latter is most likely the case. Our data indicate that mutation of residue W323 to an alanine would not only destabilize the external S1/KCNE1/pore domain interface (Figure 4D, Table 1) but also eliminate direct hydrophobic contacts which normally occur between the W323 side chain and mefenamic acid, thus facilitating drug dissociation (Figure 2D, Supplemental movie S3).

A reduced interaction with S1 could also conceivably curtail the ability of $I_{Ks}$ activators to slow dissociation of the activated complex and restore faster deactivation rates such that there is no longer an enhanced step current in the presence of the drug, as seen in mutants K41C and A44C for mefenamic acid (Figure 1A and 3A) and A44C, Y46A, W323A and Y148C for DIDS (Figure 7A). Similarly, electrostatic interactions of residue K41, which acts as a lid for the binding pocket for mefenamic acid (Supplemental movie S2), could help stabilize the protein-protein interface at the external region of the channel complex. Mutation of residue K41 to a cysteine may prevent stable $I_{Ks}$ activator binding to the channel by increasing the fluctuations of the external KCNE1 region (Figure 4C) and by reducing van der Waals contacts with the drug (Figure 2D). In simulations, mefenamic acid was seen dissociating from mutant K41C, A44C, and W323A channel complexes, and this explains why the electrophysiological effects of $I_{Ks}$ activators were prevented by the mutational substitutions at these locations (Supplemental movies S2, S3). Shifts in, and particularly reductions in KCNE1/3 association for both drugs (Table 6) that lead to loss of efficacy suggest that it is the interactions with KCNE1 and S1 that are key to maintaining the activated state. This is similar to the proposal for tight interactions between S1 and KCNE3 in maintaining constitutive activity in KCNE3-associated channels (Kasuya and Nakajo, 2022).

In close proximity to this drug-binding pocket are known gain-of function mutations in S1, S140G and V141M, which have previously been reported to slow deactivation in the presence of KCNE1 (El Harchi et al., 2010; Peng et al., 2017). As their current waveforms are qualitatively similar to those seen after exposure of WT EQ to $I_{Ks}$ activators, we propose that the same mechanism may underlie the effects of mefenamic acid, DIDS and these $I_{Ks}$ S1 mutations. In addition, we find that mutation of the neighboring residue, L142, also produces current waveforms and G-V plots which mirror those seen when WT EQ is treated with 100 µM mefenamic acid (Figure 3-figure supplement 1) or DIDS. In the cryo-EM structure of KCNQ1-KCNE3 (Sun and MacKinnon, 2020), residue L142 interacts with KCNE3, whereas V141 interacts with both KCNE3 and the pore domain. This suggests that both the V141M and L142C mutations could directly alter the interaction of the S1 domain with the pore and/or the position or movement of KCNE1. The co-evolved interface between the extracellular end of S1 and the pore domain is thought to be important for bracing the VSD, to allow efficient force transmission to the pore (Lee et al., 2009), but can also impact permeation, as the S140G and V141M mutations also enhance rubidium permeation through $I_{Ks}$ complexes (Peng et al., 2017). The importance of this S1-pore coupling to channel function is supported by mutational analyses of S1 residues (Chen et al., 2003; Hong et al., 2005; Wang et al., 2011; Campbell et al., 2013) as well as the L142C mutation examined in this study (Figure 3-figure supplement 1), which all display current waveforms with instantaneous onset. Incidentally, A300, a residue in the pore region of KCNQ1, which also interacts with both mefenamic acid and DIDS in MD simulations (Figure 2-figure supplement 2, Figure 6), has also been implicated in the same gain-of-function cleft as S140G and V141M (Smith et al., 2007). The V300T mutant has an ~-20 mV shift in V1/2 of activation and faster rates of activation (Bianchi et al., 2000), once again showing how a mutation can mirror the effects of the two activators studied here.

## Conclusion

Binding of mefenamic acid and DIDS to the extracellular end of KCNE1 and the KCNQ1 S6 and S1 helices is facilitated by a number of key residues. Residue K41 acts as a “lid” holding mefenamic acid in place, while residue W323 impacts the size of the binding pocket. Size reduction mutations of either residue destabilize mefenamic acid binding and ultimately lead to drug detachment from the channel complex. This explains why, when all four $I_{Ks}$ subunits are mutated, as in the case of K41C-EQ, EQ-W323A and A44C-EQ, little to no effect of the drug is seen. The larger drug, DIDS, interacts with many of the same residues but those deeper in the pocket appear more important than for mefenamic acid. Furthermore, the qualitative similarities between the S1 mutant channel, EQ-L142C and WT EQ in the presence of 100 µM mefenamic acid suggest that $I_{Ks}$ activators most likely cause their effects by modulating interactions between the S1 helix, pore turret, KCNE1 and the S6 helix. Upon binding, both DIDS and mefenamic acid induce conformational changes in an occult binding pocket and stabilize the S1/KCNE1/pore complex, which ultimately slows deactivation. The results indicate that this extracellular inter-subunit interface forms a generalized binding site which different drugs can access and through which induce common effects on channel activation and deactivation. The presence of such a binding site and the variable nature of $I_{Ks}$ complex composition may serve as starting points for future drug development projects targeted at discovering therapeutically-useful $I_{Ks}$ agonists.

# Materials and Methods

## Drug docking and Molecular Dynamic simulations

A model of the $I_{Ks}$ channel complex – termed pseudo-KCNE1-KCNQ1 (ps-$I_{Ks}$) – was created based on the cryogenic electron microscopy (cryo-EM) structure of KCNQ1-KCNE3 (PDB ID: 6v01) (Sun and MacKinnon, 2020). In this structure, the extracellular residues of KCNE3, R53-Y58, were substituted with homologous KCNE1 residues D39-A44. Conformational sampling was then performed for substituted residues and the lowest free energy conformations were selected for subsequent docking experiments applying a four-dimensional (4D) docking approach which accounts for the flexibility of the receptor site configuration (Bottegoni et al., 2009). A receptor ensemble was then created via another round of conformational sampling for the external part of ps-KCNE1 and its KCNQ1 neighborhood (8 Å cut-off distance) and generated conformations were used for a new docking iteration. A ligand-channel conformation with the lowest free energy was chosen from final docking iterations. The docking and conformational sampling experiments as well as *in silico* mutations described in this study were performed with ICM-pro 3.8 software (Molsoft LLC, San Diego) (Neves et al., 2012). The coordinates of selected mefenamic acid-bound ps-$I_{Ks}$ channel complexes with the lowest free energy were then used for MD. To keep the voltage-sensing domain (VSD) in its upstate conformation, we restrained the PIP2 molecule in its crystallographic position during MD simulations. Simulations were performed in a water environment with AMBER20 using a ff14SB force field for proteins and GAFF/AM1-BCC scheme for the ligand parameterization and calculation of the atomic point charges (Jakalian et al., 2002; Case et al., 2005; Maier et al., 2015). The complex was solvated with TIP3P water models and $Na^+/Cl^-$ at 100 mM concentration. The system was minimized and equilibrated during 10 ns. In all simulations described in this study the Langevin thermostat was set with collision frequency 2 $ps^{-1}$, the reference temperature 310 degrees K, and Monte Carlo barostat with reference pressure at 1 bar (Oliver et al., 1997; Wu et al., 2016). The long-range electrostatic interactions with a cut-off at 10 Å were treated with the Particle Mesh Ewald (PME) method. Bonds involving only hydrogens were constrained with the SHAKE algorithm and a 2 fs integration time step was used. The 2D diagrams in Figures 2 and 6 were generated by ICM pro software. Cut-off values are as follows: hydrophobic contacts 4.5 Å and hydrogen bond strength 0.8. Hydrogen bond strength is a unitless parameter for H-bond calculation in ICM-pro and lower values identify weaker hydrogen bonds. The values do not represent H-bond energy. The MM/PBSA and MM/GBSA methods were used to calculate the free energy of binding as described previously (Miller et al., 2012) after collection of 1000 snapshots for each run. A schematic representation of the general workflow for ps-$I_{Ks}$ model construction, drug docking and MD simulations can be found in Figure 2-figure supplement 1.

## Reagents and solutions

DIDS (Sigma-Aldrich, Mississauga, ON, Canada) was prepared as a 50 mM stock solution dissolved in 100% dimethyl sulfoxide. Stock DIDS solution was diluted in control whole-cell bath solution to obtain a final DIDS concentration of 100 μM which was perfused onto mammalian cells for whole-cell experiments. All other reagents and solutions were prepared as described (Wang et al., 2020). Mefenamic acid (Tocris Bioscience, Oakville, ON, Canada) was used at concentrations of 100 μM and 1 mM. To block $I_{Ks}$, the specific $I_{Ks}$ inhibitor HMR1556 (Tocris Bioscience, Oakville, ON, Canada) was used at a concentration of 0.3-1 μM.

## Molecular biology, cell culture and whole cell patch clamp

tsA201 transformed human embryonic kidney 293 cells were first cultured, then plated for whole-cell experiments and, finally, transfected with Lipofectamine2000 (Murray et al., 2016;

; Westhoff et al., 2019; Wang et al., 2020). All mutations were first generated using site-directed mutagenesis and Pfu turbo, then sequence confirmed. Whole-cell experiments were conducted 24-48 h post transfection. For wildtype (WT) EQ and mutant x-EQ-Y (where “x” denotes a KCNE1 mutation and “y” denotes a KCNQ1 mutation; for example, K41C-EQ, EQ-W323A), cells were transfected with a linked KCNE1 and KCNQ1 cDNA (2-3 µg was used) which assembles as a fully saturated 4:4 ratio of KCNE1 to KCNQ1. All constructs were also co-transfected with 0.8 µg of GFP to allow transfected cells to be identified. Data were obtained using an Axopatch 200B amplifier, Digidata 1440A digitizer and pCLAMP 11 software (Molecular Devices, LLC, San Jose, CA).

## Data analysis

Conductance-voltage (G-V) relationships were obtained from the normalized peak of the initial tail current (G/Gmax) and plotted against the corresponding voltage. G-V plots were fitted with a Boltzmann sigmoid equation to obtain the voltage at half-maximal activation ($V_{1/2}$) and slope (*k*) values (Tables 1 to 3). For each EQ mutant, the change in activation $V_{1/2}$ ($\Delta V_{1/2}$=$V_{1/2}$ in the presence of drug-$V_{1/2}$ control) was also determined (Figures 3E and 7E). In some cases, the foot of the G-V curve did not reach 0 (essentially in the presence of drug) and as a result, the $V_{1/2}$ was read directly from the plots. Slowing of tail current decay was used as a measure of mefenamic acid and DIDS impact on WT and mutated EQ channel complexes. Specifically, the peak to end difference currents were calculated by subtracting the minimum amplitude of the deactivating current from the peak amplitude of the deactivating current. The difference current in mefenamic acid or DIDS was normalized to the maximum control (in the absence of drug) difference current and subtracted from 1.0 to obtain the normalized response (Figures 3D and 7C).

GraphPad Prism 9 (GraphPad Software, San Diego, CA) was used to analyze all the data. Where applicable, unpaired t-test or one-way ANOVA followed by the Fisher’s least significant difference (LSD) test was used to determine statistical significance. In all figures ****, ***, **, * denotes a significance where $p<0.0001$, $p<0.001$, $p<0.01$ and $p<0.05$, respectively. All data in the figures and tables are shown as mean ± SD or SEM. Bar graphs showing mean $\Delta V_{1/2}$ (Figures 3E and 7D) were generated by calculating changes in $V_{1/2}$ induced by drug treatment vs. control in separate cells. $\Delta V_{1/2}$ values reported in the Results were calculated from the mean $V_{1/2}$ values shown in Tables 1 and 3.

# Abbreviations

- $I_{Ks}$

  delayed cardiac rectifier $K^+$ current

- LQTS

  long QT syndrome

- DIDS

  4,4’-diisothiocyano-2,2’-stilbenedisulfonic acid

- SITS

  4-acetamido-4’-isothiocyanatostilbene-2,2’-disulfonic acid

- HMR1556

(3R,4S)-(+)-N-[3-hydroxy-2,2-dimethyl-6-(4,4,4-trifluorobutoxy) chroman-4-yl]-N-methylmethanesulfonamide

- G-V

  conductance-voltage relationship

- *k*

  slope factor

- $V_{1/2}$

  voltage at half-maximal activation

- WT

  wild type

- ps-$I_{Ks}$

  pseudo-KCNE1-KCNQ1

- MD

  molecular dynamics

- MM

  molecular mechanics

- PBSA

  Poisson-Boltzmann surface area

- GBSA

  generalized Born surface area.

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## Authorship Contributions

### Participated in research design

Jodene Eldstrom, David Fedida, Vitya Vardanian, Harutyun Sahakyan, Magnus Chan, Marc Pourrier, Yundi Wang

### Conducted experiments

Magnus Chan, Harutyun Sahakyan, Daniel Sastre, Marc Pourrier, Jodene Eldstrom, Yundi Wang, Ying Dou, David Fedida

### Performed data analysis

Magnus Chan, Jodene Eldstrom, Harutyun Sahakyan, Yundi Wang

### Wrote or contributed to the writing of the manuscript

David Fedida, Harutyun Sahakyan, Jodene Eldstrom, Magnus Chan, Marc Pourrier, Yundi Wang, Vitya Vardanian

## Footnotes

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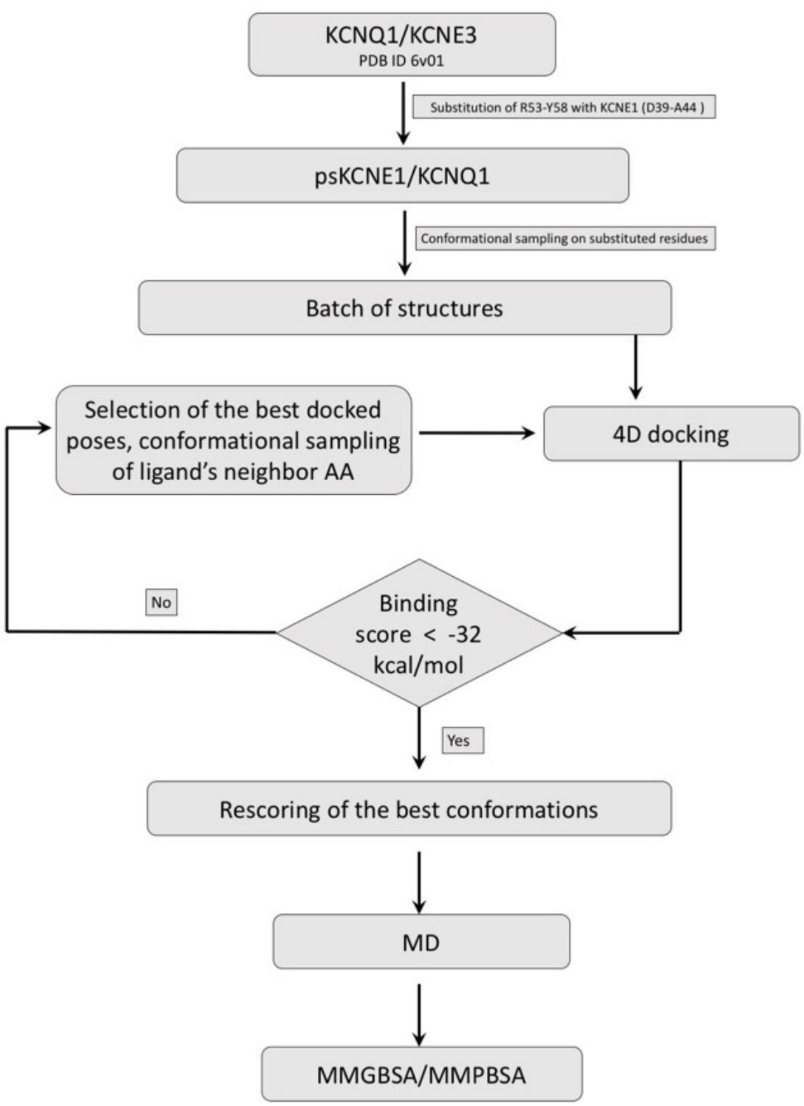



**Figure 2-figure supplement 1.**

**Drug docking and MD simulation workflow.**

(A) Schematic representation of the general workflow for construction of ps-$I_{Ks}$, Mef docking to the model ps-$I_{Ks}$ channel complex and MD simulations (See Methods).

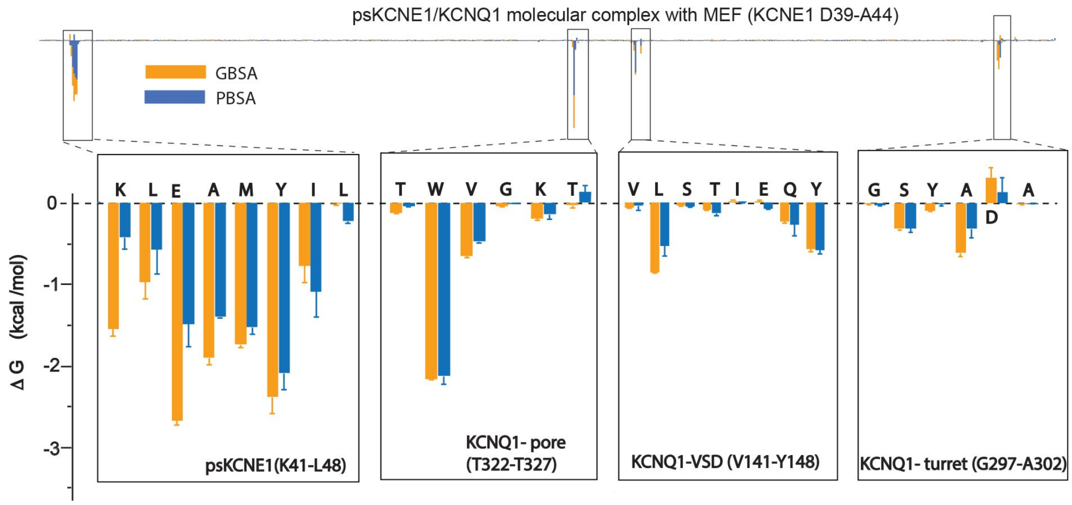



**Figure 2-figure supplement 2.**

## Energy decomposition per amino acid for mefenamic acid binding to ps-$I_{Ks}$.

Poisson-Boltzmann Surface Area (PBSA; blue) and Generalized Born Surface Area (GBSA; orange) methods were used to estimate the interaction free energy contribution of each residue in the Mef-bound ps-$I_{Ks}$ complex. The lowest interaction free energy for residues in ps-KCNE1 and selected KCNQ1 domains are shown as enlarged panels (n=3 for each point).

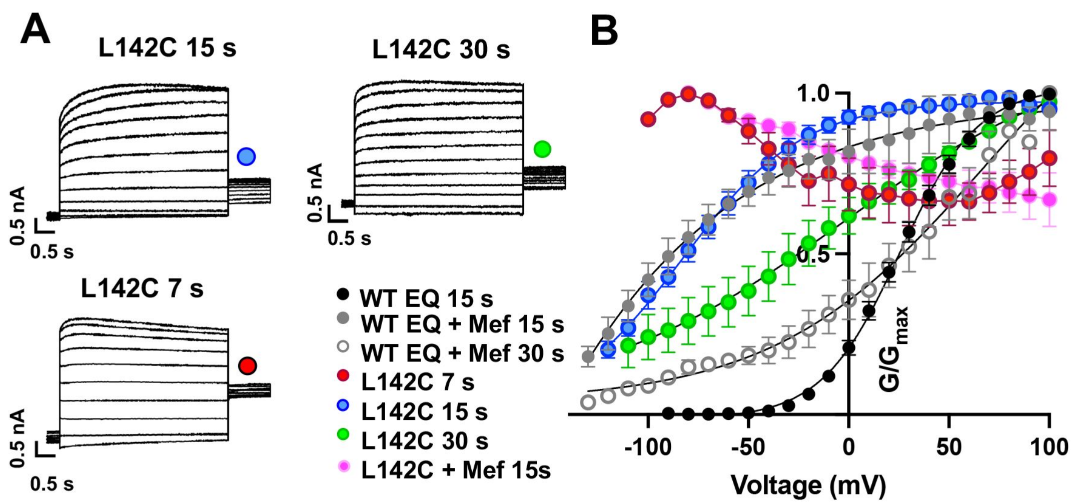



**Figure 3-figure supplement 1.**

## Augmented activation of EQ-L142C in the absence of mefenamic acid.

(A) EQ-L142C current traces in control solutions. Voltage steps were from -130 mV or higher in 10 mV steps to +100 mV for 4 s, followed by a repolarization step to -40 mV for 1 s. Holding potential was -80 mV. Interpulse interval was as indicated above, 15 s, 30 s and 7 s.

(B) G-V plot of WT EQ in control (interpulse interval 15 s: black) and in the presence of 100 μM Mef (interpulse interval 15 s: grey closed circle; 30 s: grey open circle) and EQ-L142C in control solutions (interpulse interval 7 s: red; 15 s: blue; 30 s: green). Boltzmann fits were: WT EQ 15 s control (n = 6): $V_{1/2}$ = 25.4 mV, $k$ = 19.4 mV; WT EQ 15 s Mef (n = 3): $V_{1/2}$ = -80.3 mV, $k$ = 41.3 mV; WT EQ 30 s Mef (n = 8); $V_{1/2}$ = 26.7 mV, $k$ = 66.6 mV; EQ-L142C 15 s control (n = 6) $V_{1/2}$ = -80.3 mV, $k$ = 30.0 mV; EQ-L142C 30 s control (n = 3) $V_{1/2}$ = -28.7 mV, $k$ = 62.5. Due to the dramatic change in G-V plot shape, Boltzmann fits were not possible for EQ-L142C with an interpulse interval of 7 s. All mutations are forced saturated $I_{Ks}$ channel complexes.

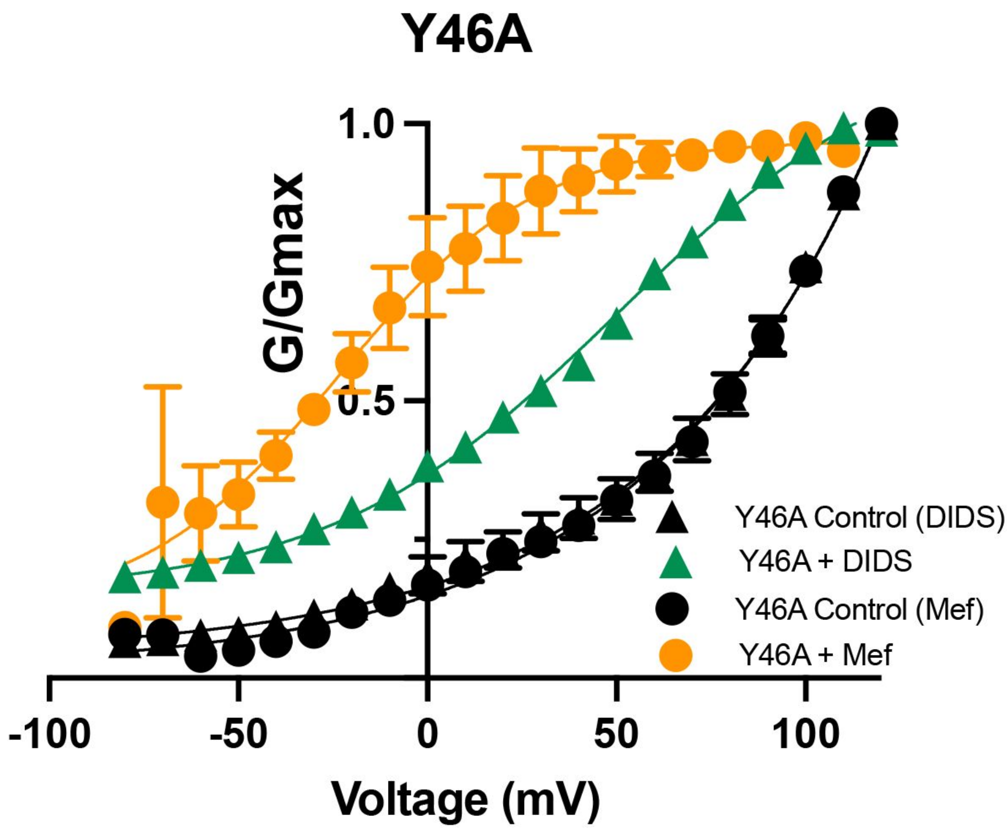
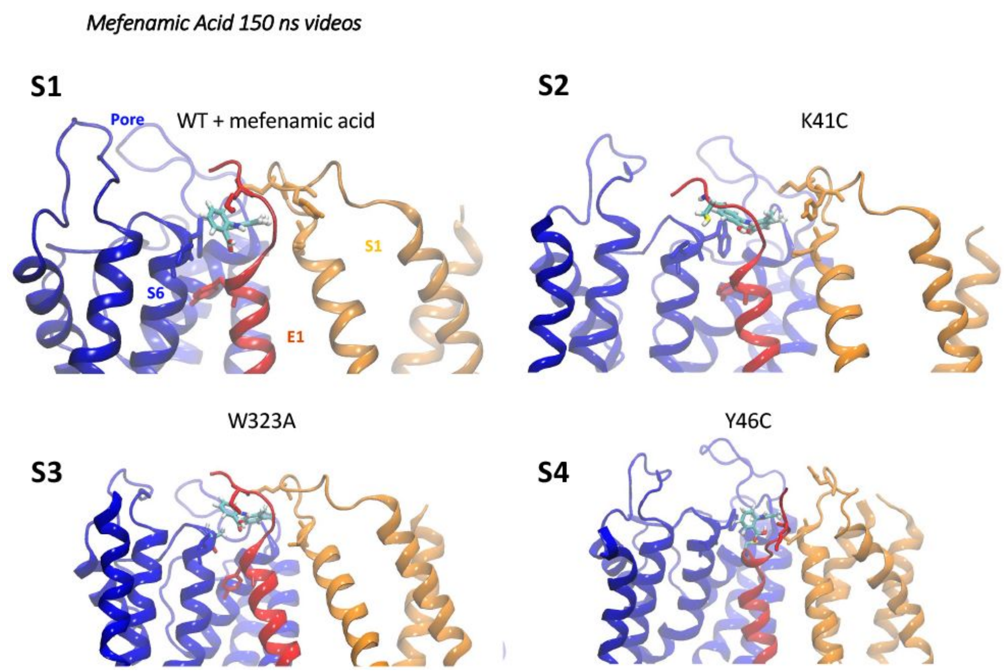



**Figure 7-figure supplement 1:**

**G-V plots for Y46A-EQ in the presence and absence of 100 mM mefenamic acid and DIDS.**

G-V plots obtained from Y46A-EQ tail currents (circles and triangles) in the absence (control: black) and presence of Mef (100 µM: orange) and DIDS (100 µM: green).

Boltzmann fits were: Y46A-EQ control (n = 3): $V_{1/2}$ = 76.4 mV, *k* = 57.3 mV; Y46A-EQ Mef (n = 3): $V_{1/2}$ = -29.8 mV, *k* = 18.9 mV; Y46A-EQ control (n = 5): $V_{1/2}$ = 75.2 mV, *k* = 59.1 mV; Y46A-EQ DIDS (n = 5): $V_{1/2}$ = 25.8 mV, *k* = 39.2 mV.




**Supplemental movies 1-4.**

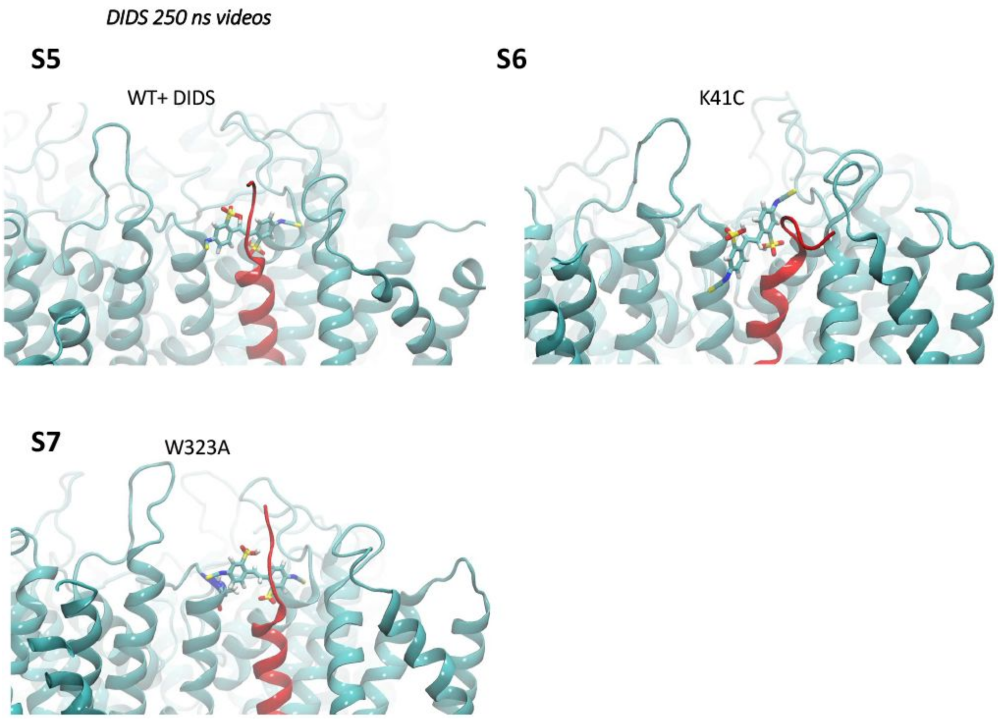
**MD simulations at the molecular level of binding of mefenamic Acid to WT EQ, and K41C-EQ,W323A-EQ, Y46C-EQ mutants. Note that videos are longer than the actual simulations, durations stated below.**

**Movie S1.** 80 ns simulation of mef binding to WT ps-KCNE1-KCNQ1.

**Movie S2.** 150 ns simulation of mef binding to K41C ps-KCNE1-KCNQ1

**Movie S3.** 110 ns simulation of mef binding toW323A ps-KCNE1-KCNQ1.

**Movie S4.** 120 ns simulation of mef binding to Y46C ps-KCNE1-KCNQ1




**Supplemental movies 5-7.**

**MD simulations at the molecular level of binding of DIDS to WT EQ, K41C-EQ, andW323A-EQ.**

**Movie S5.** 220 ns simulation of DIDS binding to WT ps-KCNE1-KCNQ1.

**Movie S6.** 250 ns simulation of DIDS binding to K41C ps-KCNE1-KCNQ1

**Movie S7.** 250 ns simulation of DIDS binding toW323A ps-KCNE1-KCNQ1.

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## Editors

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## Reviewer #1 (Public Review):

Chan et al. tried identifying the binding sites or pockets for the KCNQ1-KCNE1 activator mefenamic acid. Because the KCNQ1-KCNE1 channel is responsible for cardiac repolarization, genetic impairment of either the KCNQ1 or KCNE1 gene can cause cardiac arrhythmia. Therefore, the development of activators without side effects is highly demanded. Because the binding of mefenamic acid requires both KCNQ1 and KCNE1 subunits, the authors performed drug docking simulation by using KCNQ1-KCNE3 structural model (because this is the only available KCNQ1-KCNE structure) with substitution of the extracellular five amino acids (R53-Y58) into D39-A44 of KCNE1. That could be a limitation of the work because the binding mode of KCNE1 might differ from that of KCNE3. Still, they successfully identified some critical amino acid residues, including W323 of KCNQ1 and K41 and A44 of KCNE1. They subsequently tested these identified amino acid residues by analyzing the point mutants and confirmed that they attenuated the effects of the activator. They also examined another activator, yet structurally different DIDS, and reported that DIDS and mefenamic acid share the binding pocket, and they concluded that the extracellular region composed of S1, S6, and KCNE1 is a generic binding pocket for the IKS activators.

The data are solid and well support their conclusions, although there are a few concerns regarding the choice of mutants for analysis and data presentation.

Other comments:

1. One of the limitations of this work is that they used psKCNE1 (mostly KCNE3), not real KCNE1, as written above. It is also noted that KCNQ1-KCNE3 is in the open state. Unbinding may be facilitated in the closed state, although evaluating that in the current work is difficult.
2. According to Figure 2-figure supplement 2, some amino acid residues (S298 and A300) of the turret might be involved in the binding of mefenamic acid. On the other hand, Q147 showing a comparable delta G value to S298 and A300 was picked for mutant analysis. What are the criteria for the following electrophysiological study?
3. It is an interesting speculation that K41C and W323A stabilize the extracellular region of KCNE1 and might increase the binding efficacy of mefenamic acid. Is it also the case for DIDS? K41 may not be critical for DIDS, however.
4. Same to #2, why was the pore turret (S298-A300) not examined in Figure 7?

## Reviewer #2 (Public Review):

The voltage-gated potassium channel KCNQ1/KCNE1 (IKs) plays important physiological functions, for instance in the repolarization phase of the cardiac action potential. Loss-of-function of KCNQ1/KCNE1 is linked to disease. Hence, KCNQ1/KCNE1 is a highlighted pharmacological target and mechanistic insights into how channel modulators enhance the function of the channel is of great interest. The authors have through several previous studies provided mechanistic insights into how small-molecule activators like ML277 act on KCNQ1. However, less is known about the binding site and mechanism of action of other type of channel activators, which require KCNE1 for their effect. In this study, Chan and co-workers use molecular dynamics approaches, mutagenesis and electrophysiology to propose an overall similar binding site for the KCNQ1/KCNE1 activators mefenamic acid and DIDS, located at the extracellular interface of KCNQ1 and KCNE1. The authors propose an induced-fit model for the binding site, which critically engages residues in the N-terminus of KCNE1. Moreover, the authors discuss possible mechanisms of action of how drug binding to this site may enhance channel function.

The authors address an important question, of broad relevance to researchers in the field. The manuscript is generally well written and the text easy to follow. A strength of the work is the parallel use of experimental and simulation approaches, which enables both functional testing and mechanistic predictions and interpretations. For instance, the authors have experimentally assessed the putative relevance of a large set of residues based on simulation predictions. A limitation is that several methods need to be described in more detail to allow for evaluation of the presented data. Also, a more extensive presentation of representative data would be useful, along with discussions on the putative impact on drug effects of the diverse intrinsic properties of tested mutants.

## Reviewer #3 (Public Review):

The author is trying to identify the mefenamic (Mef) binding site and DIDS binding site on the KCNQ1 KCNE1 complex. The authors also try to identify the mechanism of interactions using electrophysiological recording, calculating V1/2 of different mutants, and looking at the instantaneous current and the tail current. The contribution of each residue within the binding pocket was analysed using GBSA and PBSA and traditional molecular dynamics simulation. The author is trying to argue that they share the same binding pocket and their mechanism of activation.

Strengths:

1. The effect of the WT channel in the presence of 100 uM Mef is very clear, and such an effect is clearly decreased with the E1-K41C and W323A mutation. The milder effect was observed with Q147C and Y148C mutants.

2. The effect of the WT channel in the presence of 100 uM DIDS is, again, very clear, and such an effect is clearly decreased with the E1-Y46C.

3. The author has indeed achieved their aim in addressing that the binding site for both DIDS and Mef are adjacent to each other and may indeed share a pocket in the S1-E1-pore pocket. This may help the field with drug development, targeting that region in the future.

Weaknesses:

1. The computational aspect of the work is rather under-sampled - Figure 2 and Figure 4. The lack of quantitative analysis on the molecular dynamic simulation studies is striking, as only a video of a single representative replica is being shown per mutant/drug. Given that the simulations shown in the video are extremely short; some video only lasts up to 80 ns. Could the author provide longer simulations in each simulation condition (at least to 500 ns or until a stable binding pose is obtained in case the ligand does not leave the binding site), at least with three replicates per each condition? If not able to extend the length of the simulations due to resources issue, then further quantitative analysis should be conducted to prove that all simulations are converged and are sufficient. Please see the rest of the quantitative analysis in other comments.

2. Given that the protein is a tetramer, at least 12 datasets could have been curated to improve the statistic. It was also unclear how frequently the frames from the simulations were taken in order to calculate the PBSA/GBSA.

3. The lack of labels on several structures is rather unhelpful (Figure 2B, 2C, 4B). The lack of clarity of the interaction map in Figures 2D and 6A.

4. The RMSF analysis is rather unclear and unlabelled thoroughly. In fact, I still don't quite understand why n = 3, given that the protein is a tetramer. If only one out of four were docked and studied, this rationale needs to be explained and accounted for in the manuscript.

5. For the condition that the ligands suppose to leave the site (K42C for Mef and Y46A for DIDS), can you please provide simulations at a sufficient length of time to show that ligand left the site over three replicates? Given that the protein is a tetramer, I would be expecting three replicates of data to have four data points from each subunit. I would be expecting distance calculation or RMSD of the ligand position in the binding site to be calculated either as a time series or as a distribution plot to show the difference between each mutant in the

ligand stability within the binding pocket. I would expect all the videos to be translatable to certain quantitative measures.

6. Given that K41 (Mef) and Y46 are very important in the coordination, could you calculate the frequency at which such residues form hydrogen bonds with the drug in the binding site? Can you also calculate the occupancy or the frequency of contact that the residues are making to the ligand (close 4-angstrom proximity etc.) and show whether those agree with the ligand interaction map obtained from ICM pro in Figure 2D?

7. Given that the author claims that both molecules share the same binding site and the mode of ligand binding seems to be very dynamic, I would expect the authors to show the distribution of the position of ligand, or space, or volume occupied by the ligand throughout multiple repeats of simulations, over sufficient sampling time that both ligand samples the same conformational space in the binding pocket. This will prove the point in the discussion - Line 463-464. "We can imagine a dynamic complex... bind/unbind from Its at a high frequency".

8. I would expect the authors to explain the significance and the importance of the PBSA/GBSA analysis as they are not reporting the same energy in several cases, especially K41 in Figure 2 - figure supplement 2. It was also questionable that Y46, which seems to have high binding energy, show no difference in the EPhys works in figure 3. These need to be commented on.

9. Can the author prove that the PBSA/GBSA analysis yielded the same average free energy throughout the MD simulation? This should be the case when the simulations are converged. The author may takes the snapshots from the first ten ns, conduct the analysis and take the average, then 50, then 100, then 250 and 500 ns. The author then hopefully expects that as the simulations get longer, the system has reached equilibrium, and the free energy obtained per residue corresponds to the ensemble average.

10. The phrase "Lowest interaction free energy fort residues in ps-KCNE1 and selected KCNQ1 domains are shown as enlarged panels (n=3 for each point)" needs further explanation. Is this from different frames? I would rather see this PBSA and GBSA calculated on every frame of the simulations, maybe at the one ns increment across 500 ns simulations, in 4 binding sites, in 3 replicas, and these are being plotted as the distribution instead of plotting the smallest number. Can you show each data point corresponding to n = 3?

11. I cannot wrap my head around what you are trying to show in Figure 2B. This could be genuinely improved with better labelling. Can you explain whether this predicted binding pose for Mef in the figure is taken from the docking or from the last frame of the simulation? Given that the binding mode seems to be quite dynamic, a single snapshot might not be very helpful. I suggest a figure describing different modes of binding. Figure 2B should be combined with figure 2C as both are not very informative.

12. Similar to the comment above, but for figure 4B. I do not understand the argument. If the author is trying to say that the pocket is closed after Mef is removed - then can you show, using MD simulation, that the pocket is openable in an apo to the state where Mef can bind? I am aware that the open pocket is generated through batches of structures through conformational sampling - but as the region is supposed to be disordered, can you show that there is a possibility of the allosteric or cryptic pocket being opened in the simulations? If not, can you show that the structure with the open pocket, when the ligand is removed, is capable of collapsing down to the structure similar to the cryo-EM structure? If none of the above work, the author might consider using PocketMiner tools to find an allosteric pocket (2023 eLife. https://doi.org/10.1038/s41467-023-36699-3) and see a possibility that the pocket exists.

13. Figure 4C - again, can you show the RMSF analysis of all four subunits leading to 12 data points? If it is too messy to plot, can you plot a mean with a standard deviation? I would say that a 1-1.5 angstroms increase in the RMSF is not a "markedly increased", as stated on line 280. I would also encourage the authors to label whether the RMSF is calculated from the backbone, side-chain or C-alpha atoms and, ideally, compare them to see where the dynamical properties are coming from.

14. In the discussion - Lines 464-467. "Slowed deactivation of the S1/KCNE1/Pore domain/drug complex ....... By stabilising the activated complex. MD simulation suggests the latter is most likely the case." Can you point out explicitly where this has been proven? If the drug really stabilised the activated complex, can you show which intermolecular interaction within E1/S1/Pore has the drug broken and re-form to strengthen the complex formation? The authors have not disproven the point on steric hindrance either. Can this be disproved by further quantitative analysis of existing unbiased equilibrium simulations?

15. Figure 4D - Can you show this RMSF analysis for all mutants you conducted in this study, such as Y46C? Can you explain the difference in F dynamics in the KCNE3 for both Figure 4C and 4D?

16. Line 477: the author suggested that K41 and Mef may stabilise the protein-protein interface at the external region of the channel complex. Can you prove that through the change in protein-protein interaction, contact is made over time on the existing MD trajectories, whether they are broken or formed? The interface from which residues help to form and stabilise the contact? If this is just a hypothesis for future study, then this has to be stated clearly.

17. The author stated on lines 305-307 that "DIDS is stabilised by its hydrophobic and vdW contacts with KCNQ1 and KCNE1 subunits as well as by two hydrogen bonds formed between the drug and ps-KCNE1 residue L42 and KCNQ1 residue Q147" Can you show, using H-bond analysis that these two hydrogen bonds really exist stably in the simulations? Can you show, using minimum distance analysis, that L42 are in the vdW radii stably and are making close contact throughout the simulations?

18. Discussion - In line 417, the author stated that the "S1 appears to pull away from the pore" and supplemented the claim with the movie. This is insufficient. The author should demonstrate distance calculation between the S1 helix and the pore, in WT and mutants, with and without the drug. This could be shown as a time series or distribution of centre-of-mass distance over time.

19. Given that all the work were done in the open state channel with PIP2 bound (PDB entry: 6v01), could the author demonstrate, either using docking, or simulations, or alignment, or space-filling models - that the ligand, both DIDS and Mef, would not be able to fit in the binding site of a closed state channel (PDB entry: 6v00). This would help illustrate the point denoted Lines 464-467. "Slowed deactivation of the S1/KCNE1/Pore domain/drug complex... By stabilising the activated complex. MD simulation suggests the latter is most likely the case."

20. I struggle with the term "normalised response" on Line 208. What is it being normalised to? Can this be put more explicitly in the text? If normalised to WT, why is WT EQ response only 0.8?

21. The author stated that the binding pose changed in one run (lines 317 to 318). Can you comment on those changes? If the pose has changed - what has it changed to? Can you run longer simulations to see if it can reverse back to the initial confirmation? Or will it leave the site completely?

22. Binding free energy of -32 kcal/mol = -134 kJ/mol. If you try to do dG = -RTlnKd, your lnKd is -52. Your Kd is e^-52, which means it will never unbind if it exists. I am aware that this is the caveat with the methodologies. But maybe these should be highlighted throughout the manuscript.

> **Author Response:**
>
> The authors would like to thank the Editors and reviewers for their careful consideration of our article and we express our appreciation for the work required by both Editors and reviewers to study and produce the detailed reviewer reports. We are pleased at the general consensus that our paper is of interest and highlights an important region of the channel for drug-protein interaction. We are also cognizant that the reviewer reports highlight areas where important revisions need to be made to our work before it can be considered fully complete. We will revise the paper according to the comments of the reviewers and submit a new version in the near future which we hope will become the version of record.